



Natural Resources
Canada

Ressources naturelles
Canada

Canadian Forest
Service

Service canadien
des forêts

Development and Structure of the Canadian Forest Fire Emissions Prediction System

Kerry Anderson
and a cast of thousands



Canada

Draft document - September 24, 2020

ABSTRACT

The Canadian Forest Fire Emissions Prediction System, or CFFEPS, is a model used to predict smoke plume development for Canada. The system consists of a fire-growth model, an emissions model, and a model to predict the penetration height of a smoke plume, based on thermodynamic principles. As a Canadian model, CFFEPS uses the Canadian Forest Fire Danger Rating System (CFFDRS), including the Canadian Fire Weather Index (FWI) System and the Canadian Forest Fire Behaviour Prediction (FBP) System.

This manual outlines the theory behind the CFFEPS model, the methodology followed to create the predictions and instructions necessary to run the C-version of the system.

INTRODUCTION

Every year Canada typically experiences 8,000 forest fires resulting in dozens of evacuations of communities due to smoke, each disrupting the lives and livelihoods of residents, their families, and their communities. Large-scale smoke events can blanket major population centres affecting hundreds of thousands of people (Anderson *et al.* 2018). Some of the most severe air quality events in Canada are due to smoke from forest fires. Smoke from wildfires contains ozone, CO₂, CO, NO_x, Volatile Organic Compounds (VOCs), and other chemical species. Ozone and particulate matter (PM₁₀ and PM_{2.5}) have serious health effects: approximately one-third of Canadians are susceptible (Stieb *et al.*, 2019). Industries and tourism are also affected by smoke.

In 2007 a workshop was held in Edmonton, which included provincial and federal agencies in charge of resource management, public health, and environment. This led to an initial strategy of testing and then implementing smoke forecasting systems in Canada. The BlueSky smoke forecasting framework (Larkin *et al.* 2009) was first introduced at the University of British Columbia in 2010. Later, Environment and Climate Change Canada (ECCC) developed the FireWork Wildfire Smoke Prediction System (Pavlovic *et al.* 2016), which has since been employed in Canada.

A critical element of all smoke forecasting systems is a methodology to calculate the amount of emissions generated from wildland fires as well as the penetration height of smoke plumes. The BlueSky system uses the Fire Emission Production Simulator (FEPS) and CONSUME 3.0, both developed by the US Forest Service (Anderson *et al.* 2004; Prichard *et al.* 2006). The model uses a modified form of the Briggs approach (Pouliot 2005) for plume rise.

The Canadian Forest Fire Emissions Prediction System (CFFEPS) is a model used to predict smoke plume development for Canada. It is designed to build on the techniques introduced in the Fire Emission Production Simulator (FEPS) and CONSUME 3.0, both developed by the US Forest Service (Anderson *et al.* 2004; Prichard *et al.* 2006). Currently, the system consists of a fire-growth model, an emissions model, and a model to predict the penetration height of a smoke plume, based on thermodynamic principles.

As a Canadian model, CFFEPS uses the Canadian Forest Fire Danger Rating System (CFFDRS), including the Canadian Fire Weather Index (FWI) System (Van Wagner 1987) and the Canadian Forest Fire Behaviour Prediction (FBP) System (Forestry Canada Fire Danger Group 1992). Operationally, the necessary inputs are collected by Natural Resources Canada's Canadian Wildland Fire Information System (CWFIS) and passed to Environment and Climate Change Canada, who run the model at the Canadian Meteorological Centre (CMC) in Montreal. There, 72-hour forecasts are created and used in their current operational version of the FireWork system.

THEORY

Smoke emissions and the energy generated from wildland fires is a problem in estimating the amount of forest fuel consumed by fire, which, in turn, is ejected into the atmosphere. Estimating the amount of fuel consumed involves estimating the mass of fuel consumed, which is a product of density and volume. Rearranged, this can be described as the product of fuel consumed per area (kg/m^2 or tonnes/ha) and area burned (m^2 or ha).

Fuel Consumed

Bulk density (gm/cm^3) is a measure of the density of forest fuel. Given the structure of forests, it is customary to break this down vertically, distinguishing the crown (the aboveground branches and foliage) from the forest floor, with the latter further broken down by depth into the litter, fermentation and humus layers (L, F and H). Fuel load is measured as the mass of available fuel per unit area (kg/m^2 or tonnes/ha).

Forest fires spread first as surface fires (Fig. 1). A surface fire burns from the top down and is measured as the depth of burn (cm). Also, the bulk density of the forest floor increases with depth. Together, they determine the surface fuel consumed. As the intensity of a surface fire increases, the likelihood of the fire jumping from the surface into the crown increases. As it does, the fire turns into a crown fire. A new source of fuel is accessed, and the fire will increase in intensity.

The FBP system of the CFFDRS provides predictions of the surface fuel consumption (SFC), the crown fuel consumption (CFC) and the sum of the two as the total fuel consumption (TFC) in kg/m^2 or tonnes/ha.

Fuel moisture is a controlling factor in fire behaviour and is tracked by the FWI system as the Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC) and Drought Code (DC). These are numeric ratings, or indices, of the moisture content of the litter and other fine fuels, of the loosely compacted organic layers of moderate depth, and of the deep, compact organic layers respectively. Their values rise as moisture content decreases.

The thin fuels of the litter layer exhibit a diurnal variation in fuel moisture. Typically, FFMC reaches a peak (minimum fuel moisture) at 5:00 PM LST. This affects the velocity, or rate of spread (ROS), of a fire. In turn, this increases the intensity and area growth of a fire. Under these

conditions, the likelihood of crown fire increases and thus the CFC typically exhibits a diurnal trend. On the other hand, the SFC is generally driven by moisture in the deeper layers and does not exhibit any diurnal variation. Since TFC is the sum of SFC and CFC, it reflects the diurnal variation of the CFC.

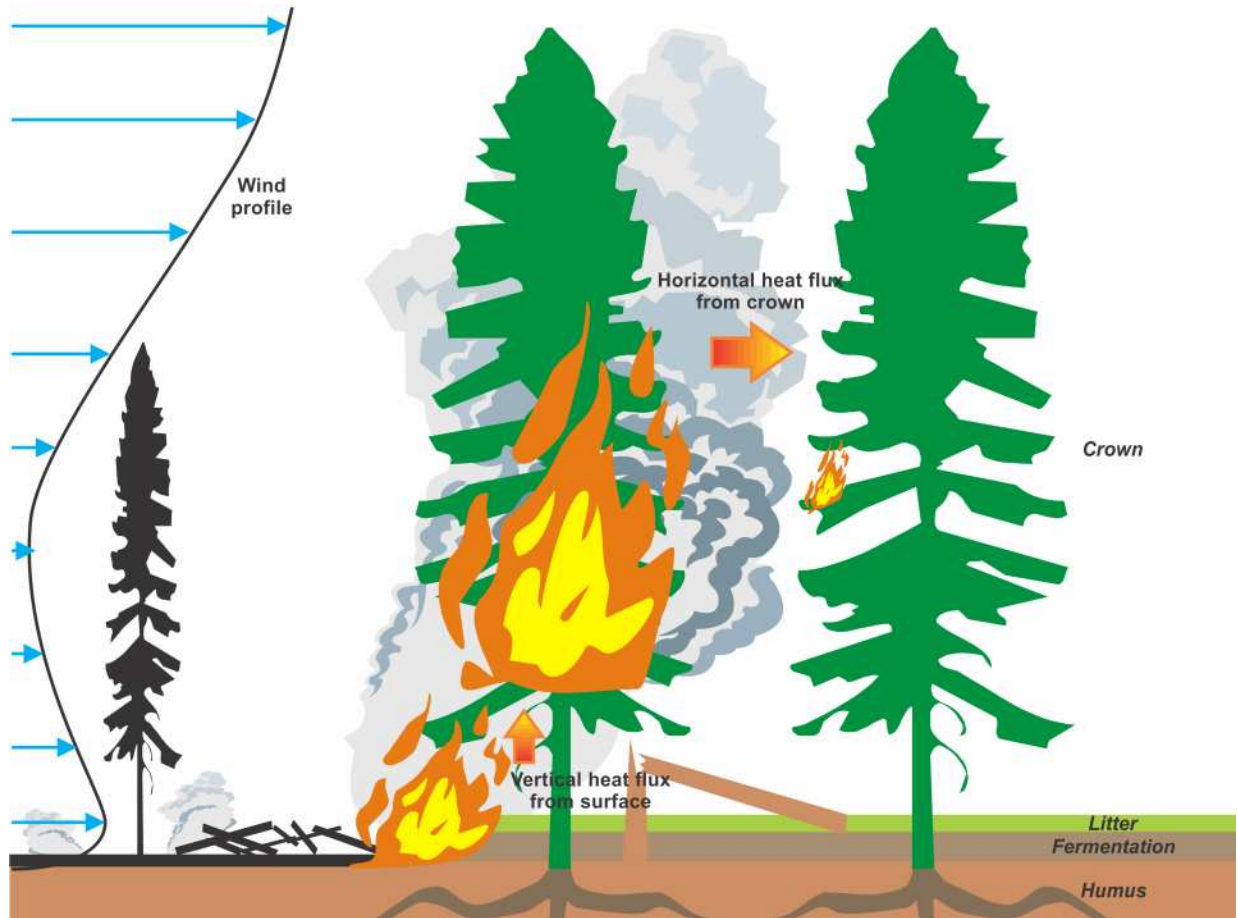


Figure 1. Fire dynamics within a forest stand. A typical wind profile associated with in-canopy flow is shown on the left with a low-level wind maximum (jet) below the principal canopy. When the surface fire reaches an intensity such that branches within the forest canopy are ignited, a crown fire may occur. Smoldering combustion can continue for hours after the principal fire front has passed.

Area Burned

The area burned by a fire is dependent on fire size, which increases geometrically over time. The rate and process of this evolution of fire size are referred to as fire growth.

Many of the calculations in the CFFEPS model depend on fire size. To determine this, the model uses a simple elliptical fire growth model to predict the area burned similarly to that used in Anderson *et al.* 2001. As shown in Figure 2, the model assumes that a fire grows in an elliptical shape: the stronger the driving wind, the higher the ellipticity. Fuel type and wind direction are held constant.

The fire-growth model uses components of the CFFDRS to drive it. The fine fuel moisture code (FFMC) is diurnally adjusted over time using the technique developed by Lawson *et al.* (1996). The rate of spread (ROS) is calculated over the course of the day and from this the area growth.

While several sophisticated fire-growth models exist that calculate fire spread over a heterogeneous field of fuels and terrain, CFFEPS does not go that far. The data needs of models such as Prometheus and FARSITE are extensive and outside the approach presented in this manual. Future attempts may be conducted to incorporate CFFEPS with these models, but for now are outside of the present scope of the model.

Note: it has been found that persistence of daily area burned provides the best measure of daily area burn rates. The elliptical growth model, as originally presented, rapidly diverges as area burned increases geometrically with time.

Emissions Rates

The CFFEPS model uses fuel consumption as calculated by the FBP system. This includes surface fuel consumption (SFC), crown fuel consumption (CFC) and total fuel consumption (TFC).

Surface fuel consumption (SFC) represents fuel consumption as the fire burns into the forest floor. The bulk density of the forest floor increases with depth, which has an impact on emissions. Table 1 shows the bulk densities used per fuel type. These values are based on the original bulk densities documented in the FBP literature and summarized in Anderson *et al.* 2000.

Table 1. Bulk densities (gm/cm³) of surface fuels per fuel type.

Fuel	0-2 cm	2-4 cm	4-6 cm	6-8 cm
C1		0.045		
C2	0.019	0.034	0.051	0.056

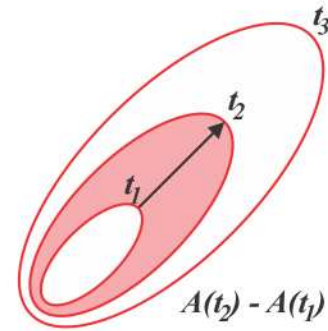


Figure 2. Elliptical fire-growth used in the CFFEPS model. The rate of spread from time t_1 to t_2 is used to calculate the forward spread distance. The grey area represents the area burned between times t_1 to t_2 .

C3	0.015	0.020	0.032	0.066
C4	0.022	0.029	0.045	0.059
C5		0.093		
C6	0.030	0.050	0.050	0.050
C7	0.100	0.100	0.050	0
D1		0.061		
M1/2	0.0265	0.071	0.0795	0.082
M3/4	0.041	0.061	0.084	0.112

Following the technique used by CONSUME, CFFEPS divides combustion into three stages: flaming, smoldering and residual combustion. Table 2 summarizes the allocation factors used by CFFEPS (referred to as reduction factors in CONSUME) to convert fuel consumption to these stages. As shown on the table, ground fuel is broken down into three layers. The litter, the upper and the lower duff layers. To estimate the amount of each layer consumed, the depth of burn must be calculated first by burning off the litter layer, then the upper duff and finally the lower duff. This is determined from the fuel type, the surface fuel consumption, and the bulk densities from Table 1. It is assumed that the first 1.2 cm describe the litter layer, the upper duff is represented by the 1.2-7 cm depth and the lower duff by 7-18 cm. Crown fraction burned is used for canopy consumption. Grass fuel consumption equals the grass fuel load.

Table 2. Allocation factors per fuel type

Fuel		Flaming	Smoldering	Residual
Ground	Litter	0.90	0.10	0
	Upper duff	0.10	0.70	0.20
	Lower duff	0	0.20	0.80
Canopy		0.94	0.06	0
Slash		0.70	0.15	0.15
Grass		0.95	0.05	0

The three combustion stages are considered to have three duration periods. Flaming combustion is considered to occur within the first 15 minutes with all consumption immediately released into the atmosphere. Smoldering and residual combustion last for several hours. Assuming the forest floor smolders at a rate of 1 cm/h, one can calculate the duration of these stages: the first half of this time is assumed to be smoldering, the second half residual smoldering.

Energy over time becomes a convolution function of fuel consumption and area burned over time

$$Q(t) = H w(t) * A(t) = \int_0^t H w(\tau) A(t - \tau) d\tau$$

where H is the heat of combustion, $w(t)$ is the fuel consumed over time t (and τ) and $A(t)$ is the area burned over time. Newly burned areas produce flaming combustion while, at the same time, old areas are still smoldering.

Table 3 describes typical emissions factors of pollutants used in CFFEPS. These are based on the average emissions factors used in CONSUME. Depending on the three stages of combustion, time series of emissions released in the atmosphere are created for each pollutant in accordance with the emissions factors and the duration of the combustion stages.

Table 3. Emissions factors (g/kg) of pollutants.

	PM	PM10	PM2.5	CO	CO2	CH4	NMHC
Flaming	11.5	7.5	6.5	45	1261	1.5	2.5
Smoldering	17	12	9.5	104.5	1142.5	5.5	5
Residual	17	12	9.5	104.5	1142.5	5.5	5

Note that the values presented in Tables 1, 2 and 3 are only examples (based on those used by the USFS). The CFFEPS program allows users to define these values through the Emissions Model CSV file, the Allocation CSV file and the Emissions CSV file (as described in the section CFFEPS Files) for Tables 1, 2 and 3 respectively.

Plume Rise

Plume rise in the CFFEPS model is based on the thermodynamic plume model developed by Anderson *et al.* (2011). This model predicts the penetration height of a plume based on an amount of energy injected into the atmosphere and a constant environmental lapse rate.

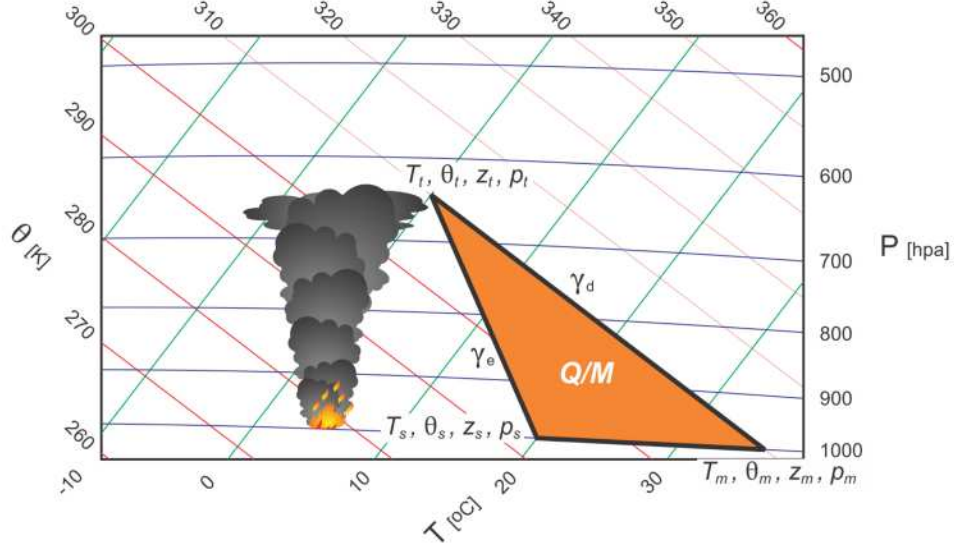


Figure 3. Representation of a smoke plume on a tephigram. The area, Q/M , indicates the energy released by the fire, which is constrained by the environmental lapse rate γ_e and the dry adiabat γ_d . Values for the temperature, potential temperature, height and pressure (T , θ , z and p) can be determined at the surface, the top and within the modified zone (subscripts s , t , and m).

The energy per unit mass, $Q/M = q$, required to modify the atmosphere can be calculated using the tephigram (Fig. 3). The quantity q can be derived from an enclosed area on the tephigram as

$$q = -c_p \oint T d \ln \theta$$

where q is the energy per unit mass [J kg^{-1}], c_p is the heat capacity of dry air [$1005 \text{ J kg}^{-1} \text{ K}^{-1}$], T is the temperature [K] and θ is the potential temperature [K]. The area enclosed within the triangular area, as shown in the figure, equals

$$q = -\frac{1}{2} c_p (T_t - T_m) \ln (\theta_m / \theta_s)$$

Energy Balance

The energy released from a wildland fire can be determined using Byram's equation

$$I = H w r$$

where I is the intensity of a fire per unit length of fire front [W m^{-1}], H is the heat of combustion [J kg^{-1}], w is the weight of the fuel consumed per unit area [kg m^{-2}] and r is the rate of spread [m s^{-1}] (Byram 1959). The heat of combustion H is a constant, typically $1.8 \times 10^7 \text{ J kg}^{-1}$ for dry wood. Following a similar format, the energy released by the fire, Q_{fire} [J] becomes

$$Q_{fire} = H w A$$

where A [m^2] is the area burned. A more rigorous representation would take the form

$$\frac{dQ_{fire}}{dt} = I L = H w r L$$

where L is length of the fire line [m].

Current fire danger rating systems provide a means of calculating the rate of spread and the fuel consumed in forest fires. Two systems, the Canadian Forest Fire Danger Rating System (CFFDRS) (Stocks *et al.* 1989) and the USDA Forest Service's National Fire-Danger Rating System (NFDRS) (Burgan 1988), provide these values. The Canadian Forest Fire Behaviour Prediction (FBP) System (Forestry Canada Fire Danger Group 1992), a component of the CFFDRS, predicts the rate of spread (ROS) [$m\ min^{-1}$] and the total fuel consumption (TFC) [$kg\ m^{-2}$]. The TFC can be used as the w term in Byram's equation to provide an estimate of the energy released into the atmosphere per square metre. Likewise, the TFC and the ROS can be used to calculate the energy released over time, which could be used to model plume development.

Not all the energy released by a forest fire enters the plume. Instead, a portion of the energy is projected ahead of the fire to heat fuels to combustion temperatures, into the ground underneath a fire (to be released later but after plume development), etc. To calculate the energy injected into the plume, an energy balance for a wildland fire was devised for CFFEPS as

$$Q_{plume} = Q_{fire} - Q_{moisture} - Q_{fuel} - Q_{radiation} - Q_{surface} - Q_{incomplete}$$

where Q_{plume} is the energy injected into the plume, Q_{fire} is the total energy of the fire, $Q_{moisture}$ is the energy lost to evaporate moisture in the fuel, Q_{fuel} is the energy required to heat the fuel to the temperature of combustion, $Q_{radiation}$ is energy lost into space away from any fuels and the plume, $Q_{surface}$ is energy injected into the ground only to be released after plume development, $Q_{incomplete}$ is energy lost due to incomplete combustion. These values are solved for in CFFEPS based on parameters from the Canadian Forest Fire Weather Index (FWI) System (Van Wagner 1987) and the Canadian Forest Fire Behaviour Prediction (FBP) System (Forestry Canada Fire Danger Group 1992).

The energy of the fire Q_{fire} (as calculated by Byram's equation) is balanced by the energy required to evaporate away fuel moisture (Q_w), the energy required to heat the fuel to combustion temperature (Q_f), the energy injected into the surface (Q_s), the energy lost due to radiation (Q_r), and the energy released into the atmosphere (Q_{plume}). Also, a certain amount of the energy of the fire is lost due to incomplete combustion (F_i), while heat from the plume is lost over time to the surrounding environment due to black body radiation (P_{BB}).

Forest fuels contain a certain water content described as fuel moisture. This water must be heated and evaporated before combustion occurs. The Canadian Forest Fire Weather Index (FWI) system (Van Wagner 1987) provides equations to calculate the water content of the litter, duff

and deeper layers as moisture contents, measured in terms of mass of water to mass of dry fuel. The total fuel consumption (TFC) value [kg m^{-2}] from the FBP system and area burned can be used to determine the dry mass of fuel, and from that the mass of water evaporated. Once the mass of water, m_w , is determined, the energy required to heat and evaporate this mass, Q_w , equals

$$Q_w = m_w(l_v + c_w\Delta T)$$

where l_v equals the latent heat of vaporization [$2.501 \times 10^6 \text{ J kg}^{-1}$], c_w is the heat capacity of water [$4.1855 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$], and ΔT is the temperature change from the pre-fire fuel temperature to boiling point (100° C).

Similarly, the dry fuel must be preheated to combustion. Again, using the product of total fuel consumption and area burned as the dry fuel mass, m_f , the energy used to preheat the fuel equals

$$Q_f = c_f m_f \Delta T$$

where c_f is the heat capacity of wood [$\sim 1.7 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$] and in this case ΔT is the temperature change from pre-fire fuel temperature to combustion ($\sim 500^\circ \text{ C}$).

While fuel must be preheated to combustion, there is an amount of fuel that is heated but does not reach combustion. In theory, the heat of the fire is isotropic and equal amounts of energy go up into the atmosphere as goes down into the ground and into other fuels. This energy is additional to the pre-combustion heating of fuel, Q_f , that contributes energy to the fire. An assumption is made that half of the energy generated by the surface fuel consumption (SFC) of the FBP system captures the energy injected into the surface.

$$Q_s = 0.5 H w_s A$$

where H is the heat of combustion, w_s is the surface fuel consumption (SFC), and A is the area burned. The choice of SFC over TFC is based on proximity. Crown Fuel Consumption (CFC where $\text{SFC} + \text{CFC} = \text{TFC}$) is assumed to not be part of the heat injected into the surface.

Byram (1959) provided an estimate that approximately 14% of the energy of the fire is emitted as radiation, Q_r

$$Q_r = 0.14 Q_{fire}$$

An alternative to Byram's value is to equate radiation lost horizontally to consider the combustion zone as a shallow cylinder. The radiation lost horizontally would then equate to the ratio of the surface area of the wall of the fire over the surface area of the top and bottom of the cylinder combined. Surface areas of the top and bottom equal the fire size, while the height of the wall of the fire equals the flame height, which is approximated by the square root of the fire intensity divided by 300 (Byram 1959)

$$Q_r = \frac{\sqrt{i/300}}{A} Q_{fire}$$

Another factor to consider is incomplete combustion. As a fire's intensity increases, the amount of available oxygen is limited, and the efficiency drops. As a result, a large amount of fuel does not burn and turns into soot, which is lifted into the atmosphere. This is apparent when the smoke from an intense fire changes from grey to black. Another variable that increases with intensity is the Crown Fraction Burned (CFB) from the FBP system. For the purposes of this model, the fraction of incomplete combustion, F_i , is set to be equal to half of the crown fraction burned (F_c)

$$F_i = 0.5F_c$$

As a convective smoke plume develops, the plume is warmer than the surrounding environment. Energy is then exchanged between the plume and the surrounding environment in the form of black body radiation. Following the Stefan-Boltzman Law, radiation is exchanged as follows

$$P_{BB} = \frac{dQ_{plume}}{dt} = \sigma T_{plume}^4 - \sigma T_{env}^4$$

where P_{BB} is the power exchange due to black body radiation [$W\ m^{-2}$], σ is the Stefan-Boltzman constant [$5.67 \times 10^{-8}\ W\ m^{-2}\ K^{-4}$], T_{plume} is the temperature of the plume [K] and T_{env} is the surrounding environmental temperature [K].

Black body radiation is measured in power and does not neatly fit into the energy balance equation. Instead, the energy held within the plume evolves over time

$$Q_{plume}(t + \Delta t) = Q_{plume}(t) + (\sigma T_{plume}^4 - \sigma T_{env}^4) \Delta t S$$

where t is time [s], Δt is the timestep [s], and S is the surface area of the plume [m^2]. Because of the vertical extent of the plume, it is necessary to use vertically averaged temperatures of the plume and of the surrounding environment.

Again, it is worth mentioning that this is an idealized model. The terms presented here are approximate and are open to change. An alternate, simpler approach would be to use a fixed percentage, P , of the total energy of the fire, Q_{fire} , being converted to energy injected into the plume (i.e. $Q_{plume} = P \times Q_{fire}$). A reasonable value for P would be between 10 and 20%.

Entrainment

The CFFEPS model provides the user with the choice of entrainment, which is represented by an entrainment angle α (Fig. 4). Essentially, the plume rises in a conical fashion expanding with height. The modified volume of the plume as captured by entrainment depends on fire size. Entrainment has a more significant role in smaller fires than in larger fires. On a large fire, the

proportional volume contributed to the plume from entrainment is minimal, while on a small fire, it is substantial (Fig. 5).

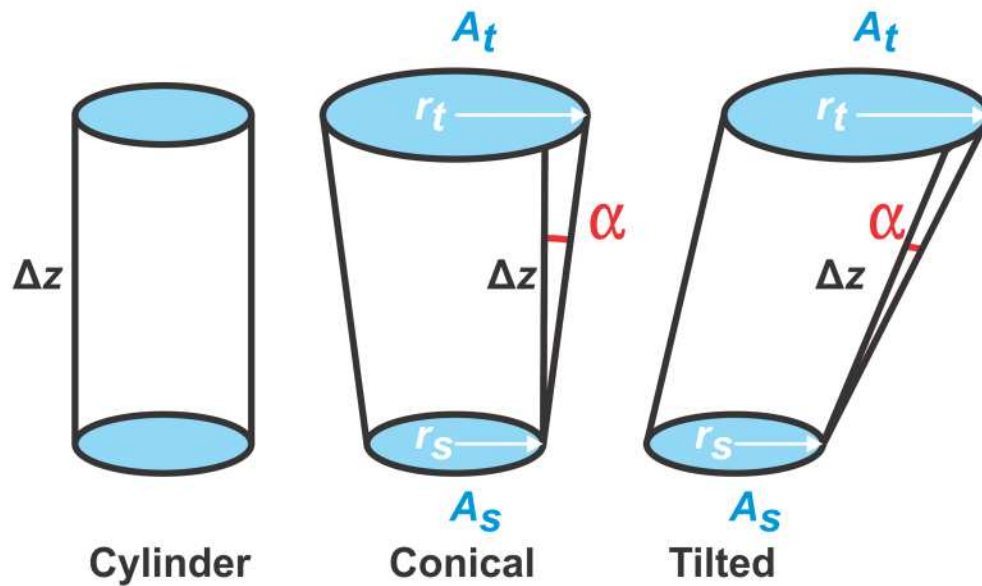


Figure 4. The effects of entrainment: without entrainment, the plume assumes a cylindrical shape; with entrainment, it assumes a conical shape following the entrainment angle α . The plume may be tilted due to winds but the plume volume and mass remains constant.

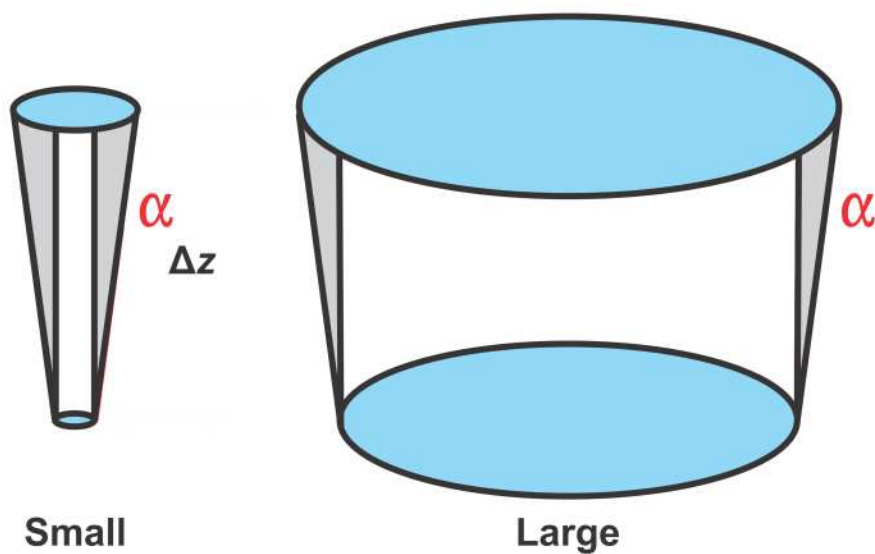


Figure 5. The contribution on entrainment (grey) is proportionally larger on the small than on the large fire.

Vertical Distribution of Emissions

As a plume rises, the density of smoke diminishes, as does the density of air. An assumption is thus made that the mixing ratio of smoke emissions to clear air is constant in the plume due to convective mixing. Given the total mass of emissions ($M_{emissions}$) and the mass of the plume (M_{plume}), the mixing ratio (r_{smoke}) can be calculated as

$$r_{smoke} = M_{emissions}/M_{plume}$$

where

$$M_{plume}/A = (p_s - p_t)/g$$

and A is the area of the fire, p_s and p_t are the pressures at the surface and the top of the plume. In the case of entrainment, the mass of the plume can be calculated using the average of the plume top area at the area of the fire.

Given the mixing ratio of smoke to clear air, the density of the smoke becomes a function of the density of air.

METHODOLOGY

Introduction

Environment and Climate Change Canada (ECCC) is mandated to include smoke emissions from forest fires in their Air Quality and Health Index (AQHI). The FIREWORK model was developed to meet this need (Pavlovic *et al.* 2016). Initially, FIREWORK used predictions based on the USFS BlueSky smoke modeling framework (Larkin *et al.* 2010) but the system is now using CFFEPS for its calculations (Chen *et al.* 2019). Over recent years, CFFEPS has gone through changes and adjustments as it develops in its operational status with ECCC.

Pre-processing

Smoke emissions produced in FIREWORK are based on satellite-detected fires, commonly referred to as hotspots, collected by the Canadian Wildland Fire Information System (CWIFS). Run by the Natural Resources Canada (NRCan) at the Northern Forestry Centre (NoFC) in Edmonton, the CWFIS collects this data several times a day and attaches forecasted fire weather and forest fuels to each hotspot.

Canadian Wildland Fire Information System

The Canadian Wildland Fire Information System (CWFIS) provides most of the data required to calculate smoke emissions and plume rise. The CWFIS is a computer-based fire information system that monitors fire activity and fire danger conditions across Canada. Daily weather observations and forecasts (up to 48 hours) are collected and used to produce fire-weather and fire-behaviour maps based on the Canadian Forest Fire Danger Rating System (CFFDRS) (Stocks *et al.* 1989). The CWFIS can be found at <http://cwfis.cfs.nrcan.gc.ca/> during the fire season.

Forecast weather for the CWFIS is provided by the Canadian Meteorological Centre of Environment Canada. Forecast data from the Global Environmental Multiscale (GEM) model (Côté *et al.* 1998) are presented in the form of SCRIBE matrices (Verret *et al.* 1997; Landry *et al.* 2005). These matrices consist of weather elements forecast from the GEM model and interpreted to specific locations within Canada. These forecast values are then adjusted using a model output statistics (MOS) approach (Wilson and Vallée 2002, 2003). The benefit of the MOS approach is that it removes any biases inherent in the GEM output. SCRIBE matrices are provided for over 800 Canadian stations or sample points on 3-h intervals for a 48-h period from the GEM model run. SCRIBE matrices are routinely available 2 hours after the GEM model run times (0000 and 1200 hours Coordinated Universal Time, UTC).

Fire detection and mapping

The Northern Forestry Centre collects hotspots detected nationally using MODIS, NOAA/AVHRR, and VIIRS satellite imagery (Quayle *et al.* 2003; Englefield *et al.* 2004). Image

pixels containing actively burning areas (hotspots) are mapped with each satellite pass (approximately every 6 hours). Fire monitoring using hotspots has various drawbacks, the most significant being the inability of the sensors to penetrate cloud. Also, geostationary satellites such as GOES are not used in this process because the resolution over Canada is judged as insufficient for this type of mapping.

Because of the resolution of the imagery used, determining hotspot sizes is not possible using this method. Instead, fire sizes at the time of detection are assigned based on annual area burned statistics per province and fuel type. In Canada, provincial and territorial agencies provide annual data on area burned. Given knowledge of the number of hotspots, an average fire size per hotspot can be calculated. Originally, this method was followed nationally but now is being conducted by fuel type provincially and territorially. Thus, for example, a hotspot detected in a C2 fuel type in Saskatchewan is assigned a fire size of 42.33 ha based on historical records. These values are updated annually by NRCan based on new, burned-area statistics. A method following ecoregion instead may be used in the future.

For each hotspot, a fire-growth simulation environment is assembled. This includes the forest fuel type, noon weather and fire weather conditions. Forest fuel type is mapped and managed in the CWFIS. In Canada, the forest protection agencies of the provinces, territories, and national parks are responsible for fire management and fuel-type mapping. Fuels are mapped from various sources, typically forest inventory, Landsat imagery, or a combination of the two. Where agency maps are not available, a national fuel type map (Nadeau *et al.* 2005) is used. The national map is based on forest inventory and Satellite Pour l'Observation de la Terre Vegetation sensors (SPOTVGT) land cover (Latifolia *et al.* 2004) and has a coarser resolution (1 km²) than the agency maps. Noon weather and fire weather indices are assigned to each hotspot based on the CWFIS. Conditions are interpolated between stations using an inverse-distance weighted approach.

All the information described to this point is collected and processed by NRCan staff at the Northern Forestry Centre in Edmonton. Information is recorded into an ASCII, comma-separated variable flat file and passed onto ECCC colleagues at the Canadian Meteorological Centre (CMC) in Montreal for further processing.

Global Environmental Multiscale (GEM) Model

Once CWFIS information is received by ECCC staff at the Canadian Meteorological Centre, 72-hour point forecasts are created for each hotspot using GEM (Fig. 8). Forecasted values include surface conditions (temperature, humidity, and wind speed) along with upper air (temperature and height) at mandatory levels (850, 700, 500, 250 mb). Each hourly forecast is included as a separate line of data. This expanded data is stored as the input file for CFFEPS.

CFFEPS Calculations

Once the necessary pre-processing has been done, the CFFEPS model is run by ECCC staff at CMC.

Fire Growth

Using the GEM forecasted hourly weather, CFFEPS calculates the fire behaviour (Fig. 6). This includes hourly calculation of the fine fuel moisture code (FFMC), rate of spread (ROS, m/min), crown fraction burned (CFB, %), surface fuel consumption (SFC, kg/m²), total fuel consumption (TFC, kg/m²), and head fire intensity (HFI, kW/m).

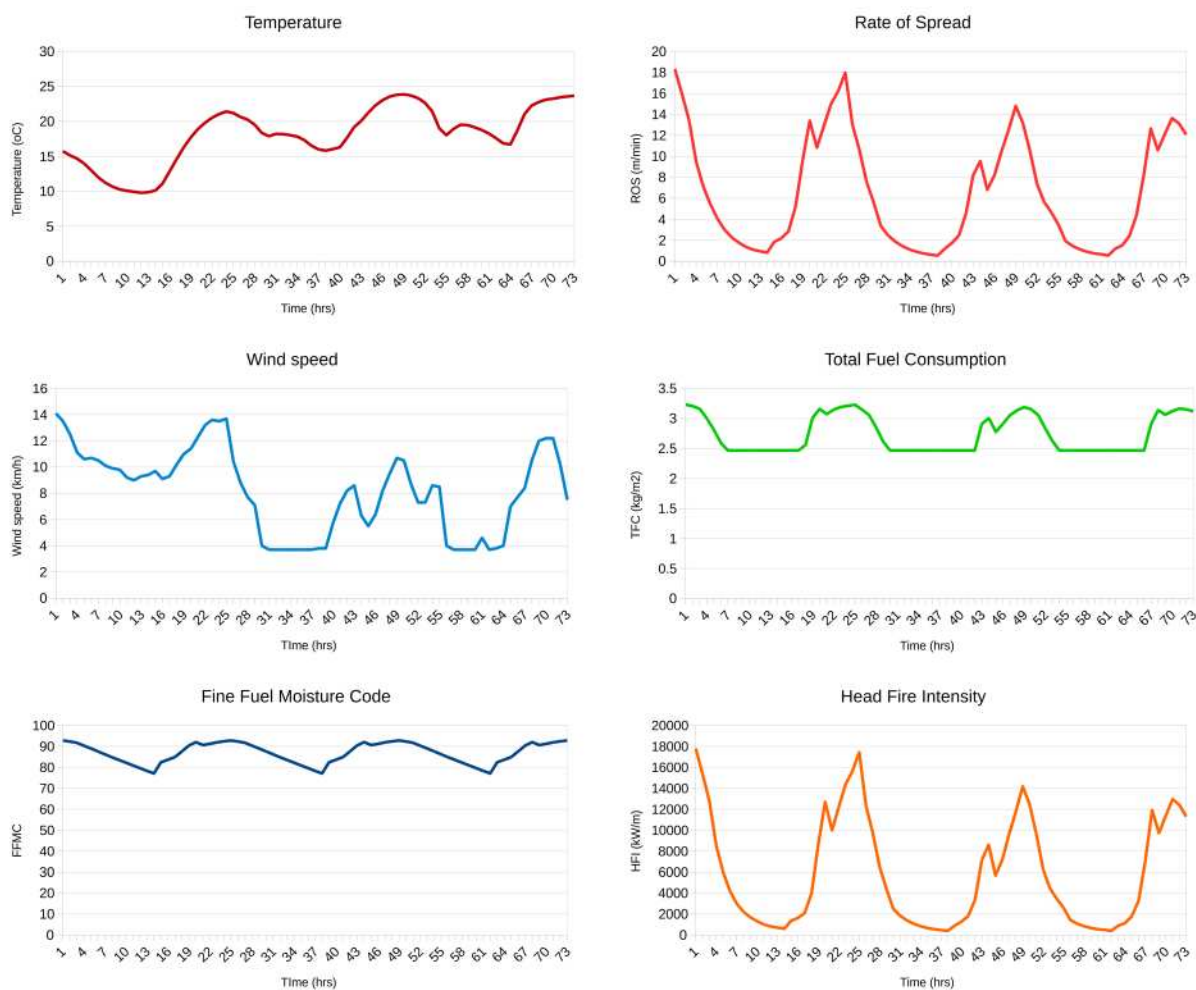


Figure 6. Modelled hourly values of temperature and wind speed from a 72 hour GEM forecasts and the resulting diurnal trends of Fine Fuel Moisture Code (FFMC), rate of spread (m/min), total fuel consumption (kg/m²) and head fire intensity (kW/m).

Note that the diurnal variation as shown in Figure 6 represent hourly changes in fire behaviour based on the hourly weather provided in the CMC forecast file. What this does not show is hourly

fire growth in terms of area burned per hour, which is predetermined by the assumption of persistence. This is addressed below.

The next step conducted by CFFEPS is the synchronization of information with the detected hotspot. The CFFEPS input file is read in one hotspot at a time (each consisting of 72 lines of hourly forecasted weather). The reported detection time (e.g., 2016-05-02 20:45) and size (e.g., *estarea* = 11.69 ha) is synchronized to the appropriate line of the forecast file. A detection time (in LST) is determined for fire behaviour prediction purposes.

From the detection time and size and the hourly fire behaviour, the diurnal growth of the fire is calculated. It has been determined that the persistence of daily fire growth (using the detected hotspot size as the daily area burned and reusing it each subsequent day) is most appropriate to create realistic overall results. From this value, the hourly distribution of the growth can be either a top-hat approach, where the daily area burned is applied evenly over a 12-hour period from 9:00 AM to 9:00 PM, or a weighted approach using a generalized diurnal trend of rate of spread over 24 hours (Fig. 7).

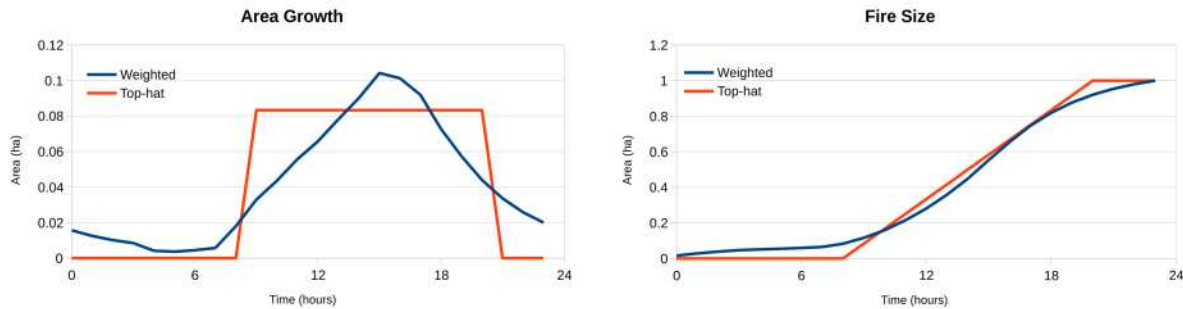


Figure 7. Area growth and fire size (accumulated area growth) over 24 hours using top-hat or weighted approach based on a detected hotspot size of 1 ha.

It is important to understand that a fire has started before the satellite passes overhead and detects it. The daily fire growth rate is thus fitted to the size at detection time. This may result in undetected growth on previous days (Fig 8).

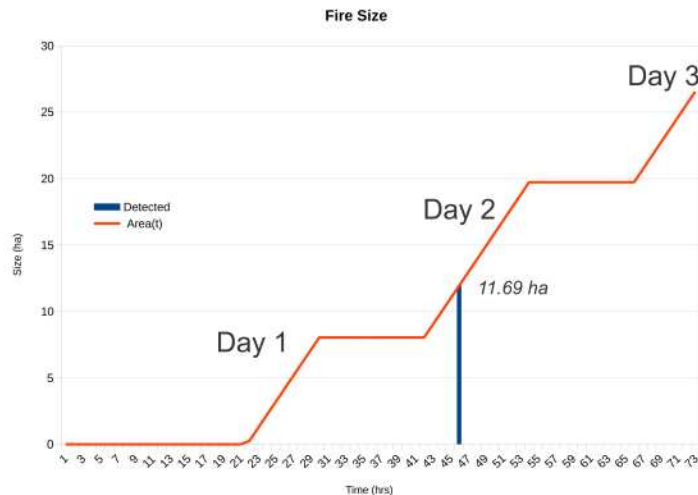


Figure 8. Fire growth using top-hat approach synchronized with the detection time (2045 UTC, 1145 LST) and size (11.69 ha). Following persistence, the fire is assumed to burn 11.69 ha per day. This results in the fire burning 9.0 ha on the day prior to detection.

Diurnal Modelling

Version 2.09 introduced a method to incorporate hourly forecasted weather into fire behaviour and diurnal modelling. The objective of this work was to find an acceptable scheme to include hourly weather in the CFFEPS calculations.

The Fine Fuel Moisture Code (FFMC) is one of the three fuel moisture indices in the Canadian Forest Fire Weather Index (FWI) System (Van Wagner 1987). It is an indicator of the moisture content within the litter layer of the forest floor, typically the top 1 to 2 centimetres. Normally, all FWI indices are calculated at noon and estimate conditions at peak burning conditions, typically 17:00 but the FFMC has a significant diurnal trend. In turn, this affects the ISI and FWI.

In 1977, Van Wagner (CVW) documented a method to calculate the hourly FFMC using hourly weather input. These hourly calculations involved a simple adjustment of the drying term in the original equations. In 1982, Alexander produced tables to estimate the hourly FFMC based on the noon value alone. These tables removed the need for hourly weather data. In 1996, Lawson *et al.* (BDL) expanded these tables and provided equations to estimate the diurnally adjusted FFMC based on the previous day's noon value. Based on the previous day's noon weather, Lawson's table cannot provide the detailed variation of the FFMC using hourly weather.

Van Wagner's model was tested against hourly-measured fuel moisture at Petawawa under a closed jack pine canopy. Adjustments to the hourly response factor were made based on these measurements. A similar study was conducted in Prince George by Lawson in an open lodgepole pine stand. Lawson's tabular values were created using this data. Later observations such as at the International Crown Fire Modelling Experiment (ICFME, Beck and Armitage 2002) and then fire growth model predictions (Anderson 2009) began to show discrepancies with the two models.

Van Wagner's model did not show the diurnal variations observed, likely explained by stand closure.

The Equilibrium Moisture Content (EMC, Anderson 2009) provides an alternative to CVWs equations, where response time is zero. Recovery is therefore instantaneous. Additionally, EMC is based on the current weather and therefore avoids the need to track past FFMFC values. Shortcomings of the EMC is that it has no memory of past conditions and that the recovery after rainfall cannot be incorporated into the model. The EMC closely resembles the diurnal variation observed in Lawson's tables (Fig. 9) and can serve as an approximation of the diurnal variation under Lawson's conditions.

Figure 9. Typical variation of FFMFC following three diurnal modelling techniques.

Prior to version 2.09, CFFEPS used the BDL approach alone to calculate hourly FFMFC. In turn, hourly FFMFC and wind affected the ISI, which then affected the fire behaviour values of ROS, CFB, CFL and TFC affecting the amount of emissions. But fire growth and area burned followed a predetermined diurnal trend (top-hat or weighted, Fig 7.) and hence was not affected by hourly weather. Because of the limitations of persistence (i.e. daily area burned remains constant), fire growth cannot be included directly, but the new approach adjusts the diurnal weighting scheme from the predetermined pattern to one that is dynamic and dependent on the hourly weather. The new weighting scheme uses a weighted moving average with the weighting scheme adjusted to ratio of the hourly ROS over the sum of the ROS of the past 24 hours.

$$Weight_{WMA}(t) = ROS(t) / \sum_{\tau=t-24}^t ROS(\tau)$$

The 24-hour weighted moving average (WMA) approach was tested against the original weighted scheme (Fig. 10) and other strategies under three varying scenarios including variable winds and rain. The WMA approach produced the desired hourly variability while still being consistent (minimal bias) with the original weighted scheme (to not invalidate earlier CFFEPS studies). Results were reliable with minimal noise compared to those using hourly alone. The new approach also allows for the use of CVW, BDL or EMC techniques.

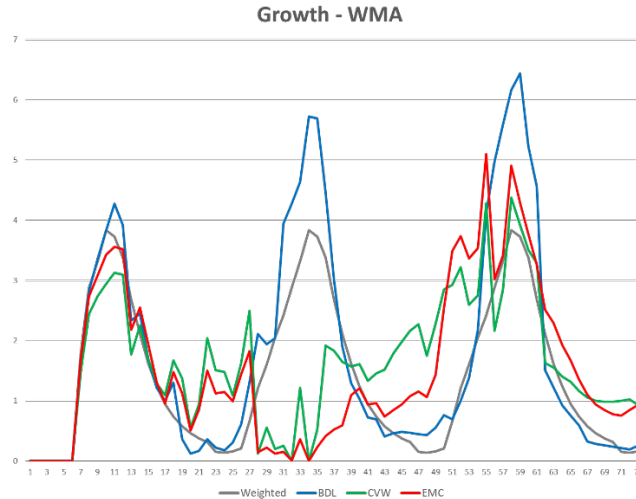


Figure 10. Growth based on weighted moving average (WMA) approach following various diurnal options (BDL, CVW, EMC). The original weighted approach shown in grey.

Fire-Growth Modelling

Version 2.10 introduces spatial fire-growth modelling into CFFEPS. This represents a significant departure from the previous versions based on observed hotspots. The new spatial fire-growth uses observed hotspots to build an ignition grid and then spreads the fire through a predetermined fuels grid from burned cell to adjacent unburned cells.

The ignition grid represents a regular grid showing the time at which the center of each cell is ignited. The ignition grid is used to initiate the fire-growth model (ignition times represent observed times of ignition prior to modelling), as well as present the fire-growth output (ignition times represent times of ignition predicted by the growth model). The starting ignition grid may be externally generated and provided to CFFEPS or it can be generated internally based on the latitude-longitude, estimated area (*estarea*) and the detection time of each hotspot.

Additional requirements for the fire-growth model include a spatial representation of the fuels, topography and forecasted weather, the later including surface and upper air. As presently constructed, all grids must conform in terms of spatial scale, alignment, and orientation.

Given the weather, fuels and topography at a given time and location, the rate of spread of a forest fire can be calculated. Systematically applying these calculations for all points currently burning along a fire perimeter enables the prediction of fire growth across the landscape; hence, fire-growth modelling.

The fire-growth model in CFFEPS uses deterministic, eight-point or sixteen-point propagation routines to estimate the time of ignition of each cell (Kourtz *et al.* 1977). Fire spread from one cell to the next is calculated through a series of one-minute time steps, allowing for temporal variation while limiting spread to the cells in question.

The directional component of spread is calculated assuming elliptical fire growth. With the ignition point at one of the foci, the directional component of the rate of spread becomes

$$r(\theta) = \frac{(1 - \sqrt{1 - 1/LB^2}) r_h}{1 - \sqrt{1 - 1/LB^2} \cos(\theta)}$$

where θ is the departure from the wind direction, $r(\theta)$ is the rate of spread with respect to θ , r_h is the head fire rate of spread and LB is the length to breadth ratio (a factor controlling fire shape, primarily dependent on the wind speed). Figure 11 illustrates the method used in fire-growth.

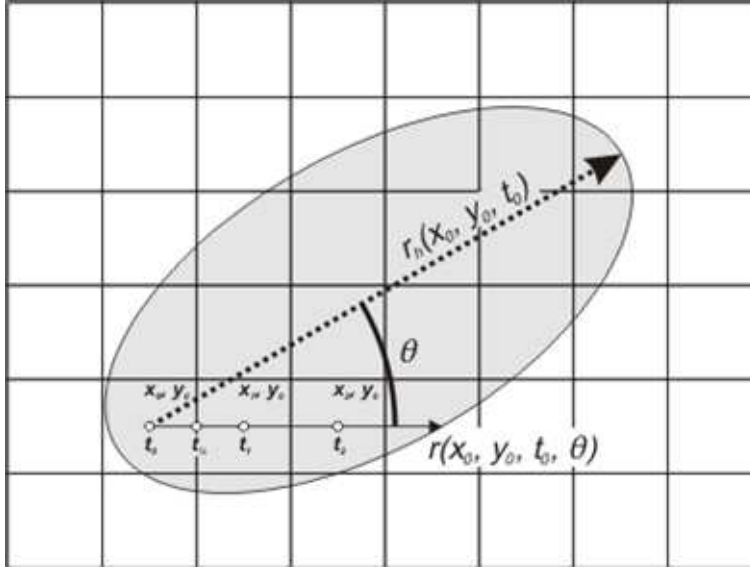


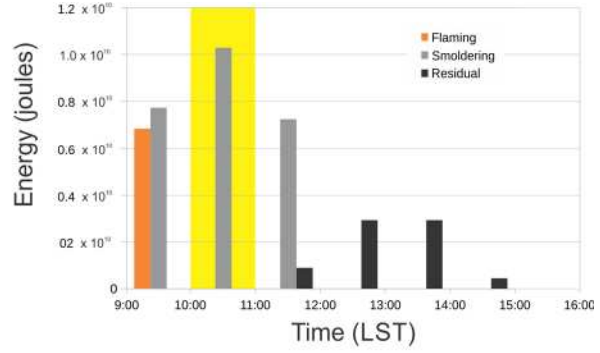
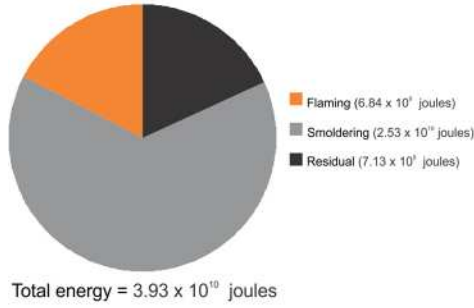
Figure 11. Illustration of the fire propagation model for a fire starting at cell (x_0, y_0) with a length-to-breadth ratio, LB (x_0, y_0, t_0) , of 2. With a wind from 240° , the head-fire rate of spread, $r_h(x_0, y_0, t_0)$, is directed toward 60° , defining the major axis of the elliptical growth. The value $r_h(x_0, y_0, t_0)$ is derived from the FBP system and depends on the weather at time t_0 , and the fuel type and topography within cell (x_0, y_0) . The rate of spread towards the east at time t_0 , $r(x_0, y_0, t_0, \theta)$, is derived using a 30° departure from the head fire rate of spread (θ) . The time when the fire exits cell (x_0, y_0) is $t_{1/2} = t_0 + [\text{cell size}/2] / r(x_0, y_0, t_0, \theta)$. The process is then repeated to move the fire to the center of cell (x_1, y_0) using $r(x_1, y_0, t_{1/2}, \theta)$, based on the fuel and topography in cell (x_1, y_0) and the weather at time $t_{1/2}$, to derive t_1 , the

ignition time of cell (x_1, y_0) . This process is repeated to calculate t_2 and so on, thus building a grid of ignition times. By using discrete time steps (minutes) and tracking the distance travelled in partial-cells, diurnal variations in the rate of spread can be captured easily.

Energy over Time

Once the hourly growth is established, CFFEPS calculates fuel consumption by depth and then flaming, smoldering and residual times for energy (and later emissions) calculations. The fraction of energy released during to the three stages is based on the USFS CONSUME 3.0 model, bulk densities from the Canadian FBP system and additional expert opinion where required (Table 2).

Fuel consumption from the current hour's fire growth is thus spread out over time. The flaming stage is assumed to occur in the first 15 minutes of combustion. Afterwards, a fire is assumed to burn into the forest floor at a rate of 1 cm/h. Knowing the depth of burn (based on surface fuel consumption and bulk density), the burn time can be calculated. Half of this time is assigned to the smoldering stage, the second half to the residual stage.



+

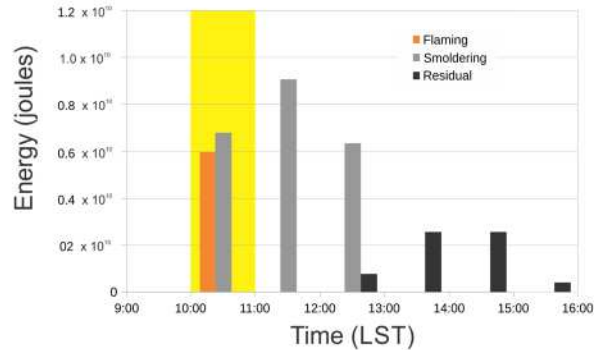
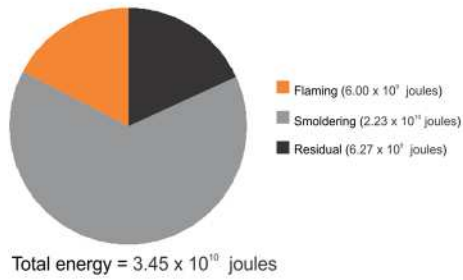


Figure 12. An example of hourly energy released (over the first two hours). During the first hour (between 9:00 and 10:00), a fire grows 0.555 ha. The energy emitted from this area burned (3.93×10^{10} joules) is spread over the next 5.15 hours (beginning at 9:00 LST) with flaming lasting 15 minutes (0.25 hours), while smoldering and residual last 2.45 hours (based on a 4.9 cm depth of burn and a smoldering rate of 1 cm/hour). During the second hour (between 10:00 and 11:00), the fire grows 0.451 ha to a total size of 1.001 ha. The energy emitted from this area burned (3.45×10^{10} joules) is also spread over the next 5.15 hours (beginning at 10:00 LST). This energy is then added to the energy released from the first area burned between 10:00 and 11:00 LST (shown in yellow). As the model steps through each hourly time step, the energies associated with each hourly area burned are distributed over time and added.

Energy values are then calculated over time (Fig. 12) as the convolution function

$$Q(t) = H w(t) * A(t) = \int_0^t H w(t) A(t - \tau) d\tau$$

Where H is the heat of combustion, $w(t)$ is the fuel consumed over time t and $A(t)$ is the area burned over time.

Note that because of the top-hat approach used in this example for fire growth, there is no energy released at night during certain hours (after the residual smoldering has finished), plume rise is turned off. Also, smoldering and residual is no longer limited to the bottom layer (as done in BlueSky).

Plume Rise

Hourly energy values are now used to calculate plume rise. CFFEPS uses several methods to calculate plume rise, which have been developed over the years. These are included as C and FORTRAN subroutines in CFFEPS.

PlumeCalc

The routine *PlumeCalc* represents the first attempt at addressing the plume rise problem. It assumes a plume will rise in the atmosphere heating a column of air above the fire from a given environmental lapse rate to that of a dry adiabat. The height of the plume is the point at which the two lapse rates cross.

Figure 3 shows the method used in the calculation, as presented on a tephigram. The energy of the fire injected into the atmosphere Q over the mass of the plume M creates an area on the tephigram Q/M (shown in orange). This heats the environmental lapse rate γ_e to the dry adiabat γ_d . The top of the plume z_t is the point at which the temperature T_t resulting from dry adiabatic lift is in equilibrium with the environment.

The *PlumeCalc* routine provides a simple method to calculate an exact answer to the plume height. It was implemented in the early stages of CFFEPS. A drawback to this approach is that the atmosphere must be represented with a single, environmental lapse rate, which raises the question of what level best represents the atmosphere. To determine this, CFFEPS follows a simple decision tree: the lapse rate from the surface to 850mb is first used; if the predicted plume is over 2000 m, the surface to 700mb is then used; if this predicted plume is over 4000m and surface to 500mb is used.

PlumeCalcDry

The routine *PlumeCalcDry* is a more detailed solution to the plume rise problem. This routine uses all the information in a sounding, predicting the energy required to lift the plume through multiple layers of buoyancy.

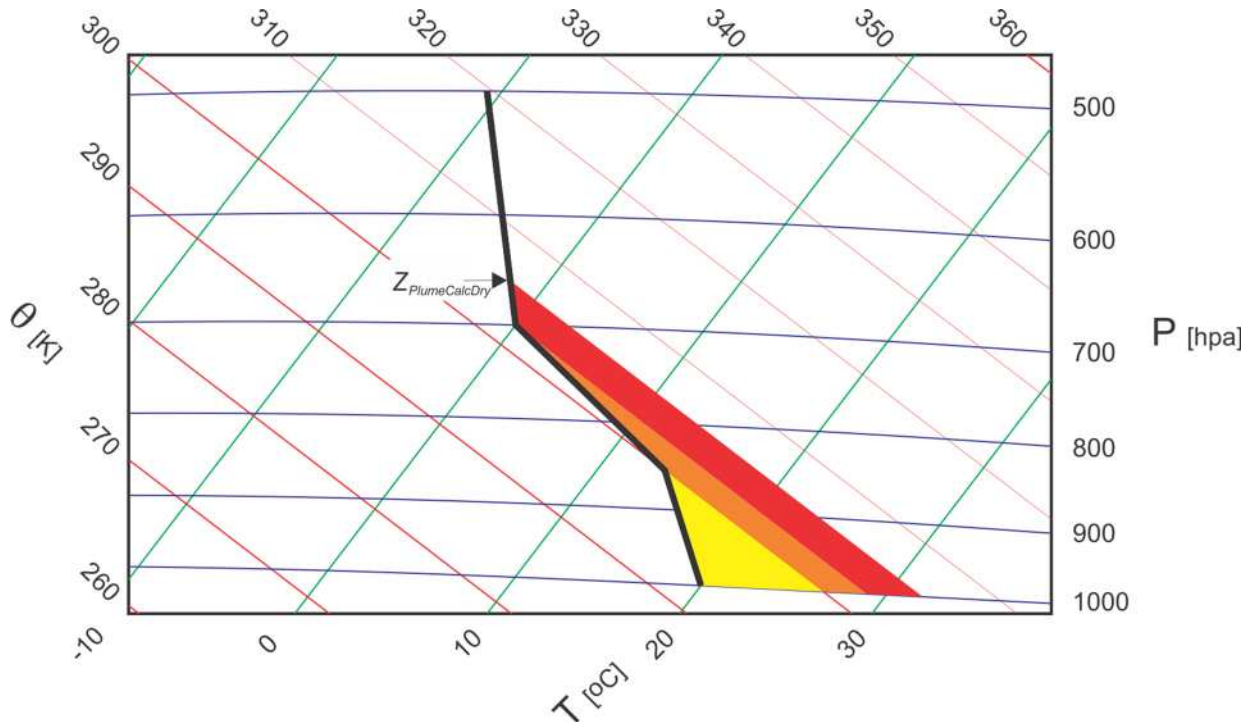


Figure 13. Thermodynamic depiction of the method used in PlumeCalcDry.

PlumeCalcDry calculates the energy required to lift a parcel through each successive layer. As shown in Figure 13, *PlumeCalcDry* calculates the energy required to heat the first layer (yellow) to a dry adiabat, followed by successive layers (orange and red). It eventually reaches a point where the cumulative energy required exceeds the energy of the fire. At this point the method reverses to a downward direction through the top layer to converge on the final answer using a bisectional approach.

PlumeCalcWet

The routine *PlumeCalcWet* takes *PlumeCalcDry* one step further, capturing condensation above the lifted condensation level (*LCL*). The *LCL* is calculated based on the mixing ratio of water to dry air r (Fig. 13) and the resulting dew point temperature T_d at the surface using the following equation

$$LCL = 120 (T - T_d)$$

Where *LCL* is the lifted condensation level [m], and T and T_d are the surface and the dew point temperatures [°C]. Figure 14 shows the additional energy (blue) added due to condensation. This energy is added to the energy of the fire to determine the final height of the plume.

It is worth noting that as the column is heated due to the fire, the *LCL* rises as the warmer plume can now hold more moisture.

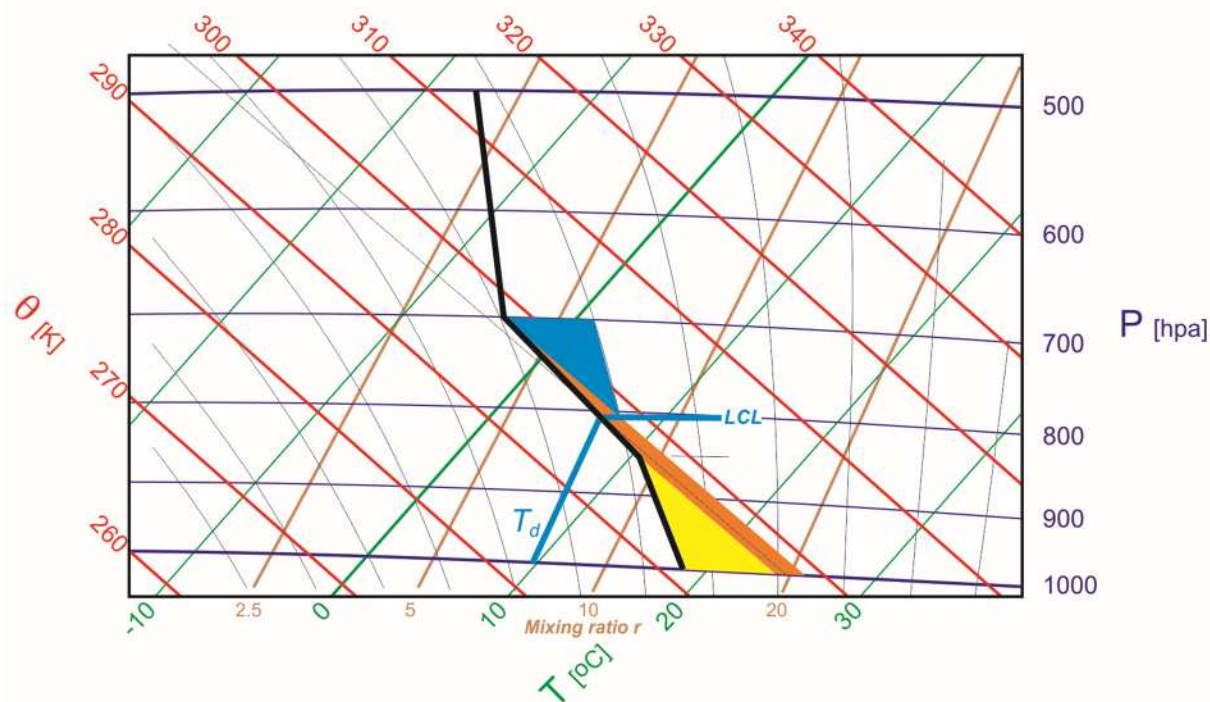
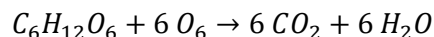


Figure 14. Thermodynamic depiction of the method used in PlumeCalcWet.

PlumeCalcWetter

A consideration now included in *PlumeCalcWet* routine incorporates the addition of moisture from evaporated fuel moisture as well as from combustion.

Fuel moisture estimates for the litter, fermentation and humus layers of the forest floor are calculated from equations used by the Fine Fuel Moisture Code, Duff Moisture Code and Drought Code (FFMC, DMC, DC) of the Canadian Forest Fire Danger Rating System. Water released from combustion is based on the combustion of simple glucose (a reasonable substitute for more complicated and variable composition in wood):



Thus for 1 kg of fuel consumed, approximately 0.60 kg of water is released.

Prior to calling *PlumeCalcWet*, the amount of water released due to fuel consumption is calculated. This follows the same processes involved in calculating hourly energy and emissions rates (e.g. PM 2.5, etc) in CFFEPS. Once hourly masses are calculated, they are introduced into the *PlumeCalcWet* calculations. As the calculations move up from the surface through the atmosphere, the mass of water is distributed to the current calculated height to create a modified the mixing ratio r and new LCL levels. Note that the mass of water from combustion is fixed and as the process steps vertically upward, this mass is stretched through a higher plume thus its contribution to plume rise is lessened.

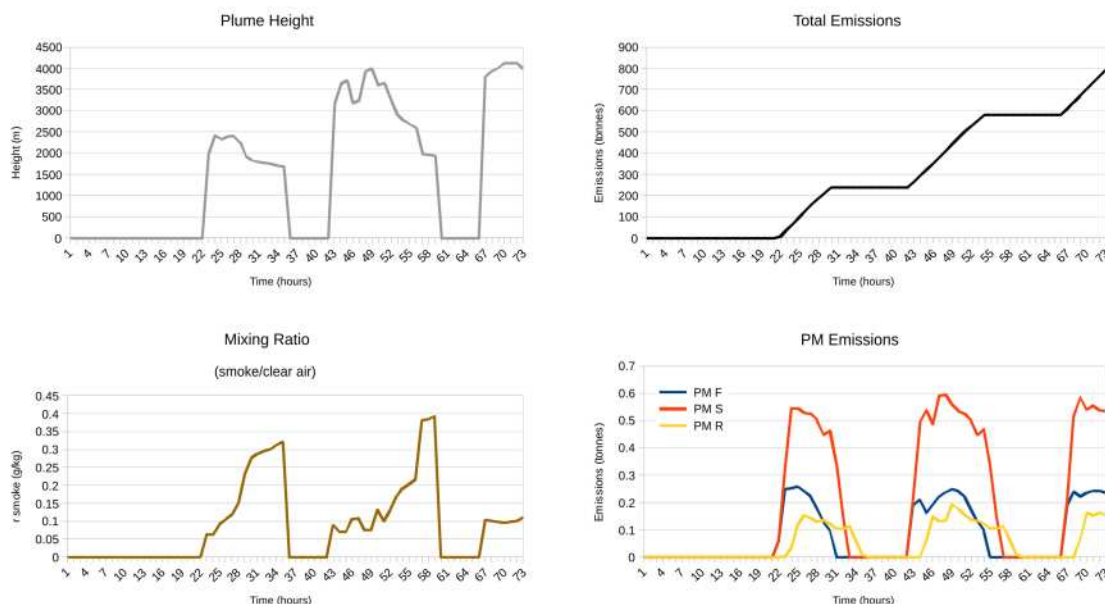


Figure 15. Examples of plume rise (m), total emissions (tonnes), mixing ratio (g/kg) and particulate matter (PM) over time.

Emissions per Species

The CFFEPS program manages emissions per species. The user defines the emissions per species as related to the total fuel consumption through the emissions CSV file and the emissions model CSV file (as described in the section CFFEPS Files). This is controlled through each stage of combustion (flaming, smoldering, residual) similar to the method presented for energy. By calculating the total fuel consumption (tonnes/h), emissions per species are calculated (tonnes/h) (Fig. 15).

Emissions are controlled through a file (e.g., emissions.csv) that contains all the emissions rates per species per stage of combustion in grams per kilogram of fuel consumption as defined by ECCC. Emissions factors can be defined by input fuel type referenced by the field emodel, which is linked to a file (e.g., emodel.csv). Actual files and file names are controlled in the CFFEPS input file (e.g., CFFEPS.INI). Currently, ECCC uses one set of emissions factors for all fuels.

Post-processing

As CFFEPS runs, it creates an output file. This output file contains the same information, line by line, as the input file but has predictions appended to it. These predictions include energy, total emissions, plume rise, and emissions by species and separated by flaming, smoldering and residual stages. This information is passed to GEMMACH for further modelling processes.

CFFEPS PROGRAM

Software has been developed to run the Canadian Forest Fire Emissions Prediction System (CFFEPS) for Environment and Climate Change Canada. This is contained in the single, executable program, **CFFEPS.EXE**, that calculates fire-growth modelling, energy balance, plume rise and fire emissions.

Running CFFEPS

CFFEPS.EXE is run from the command line as follows.

```
> CFFEPS [inputfile]
```

Where inputfile is the input file name, the default input filename is **infile.dat** in the current directory.

Input

Input is passed to **CFFEPS.EXE** through an input file. The file reads words from the file. These words are separated by spaces or tabs. A typical ECCC input file is as follows

csvfilename	C:\path\cmc2016050200_out250mb.csv
emissions	C:\path\emissions.csv
emodels	C:\path\emodels.csv
method	cmc
shape	weighted

The program looks for key words (such as method) and when it finds a keyword, it reads the following word as the data associated with the keyword (in this case cmc for the method).

Keywords are case insensitive and often the first three or four letters of the keyword are used for keyword recognition. Subsequent letters are recommended to make the input file more readable to the user but are not required.

The file must end with a Line Feed <LF> or the last line may not be read successfully.

Additional information may be included in the input file (for example a fire number or a weather station name). As these are not recognized keywords, the program will overlook them and move to the next word.

Keywords

The **CFFEPS.EXE** program was originally a robust, dynamic program designed to model plume development under a variety of conditions and scenarios. Such an approach is beyond the scope

of the current project with ECCC and so the inputs to **CFFEPS.EXE** are focussed specifically on ECCC needs. As a result, the keywords that drive the model are more limited.

Default values are show in square brackets (e.g., [default])

ALPha or ENTrainment	the entrainment half angle in degrees on the edge of the plume (default value based on personal observations of prescribed fires) [12.]
ALLOcation or REDUction	The filename (including path) of the Allocation CSV file [allocation.csv].
CONsumption or FC	Fuel consumption method (experimental) based on spatial fuel information. There are 5 choices: • 1 uses the original FBP equations and default Surface Fuel Load (SFL) values • 2 uses spatial fuel load values for surface fuels, fine fuels, and woody fuels (SFL, FFL, WFL) • 3 uses spatial values but with the original FBP's default SFL as a maximum SFC value • 4 uses specific Forest Floor Fuel Consumption (FFFC) for the SFC (de Groot et al (2009) • 5 uses the general, non fuel-specific equation for FFFC
CSVFilename	The filename (including path) of the CMC GEM Forecast CSV file [input.csv].
DIUrnal	diurnal calculation options • BDL – Lawson <i>et al.</i> 1996 • CVW – Van Wagner 1977 • EMC – Anderson 2009 • OFF [EMC]
DIURNALFile or HOURLy	The filename (including path) of the Diurnal Trend CSV file [diurnal.csv]
EMISsions	The filename (including path) of the Emissions CSV file [emissions.csv]
EMODEls	The filename (including path) of the Emissions Model CSV file [emodel.csv]

METHod	The method followed by the CFFEPS program. For ECCC, the method used is CMC. If the keyword consists of CMC plus more characters (e.g. CMCnew or CMC2), the program will read in the new, extended CSV hotspot input file that uses more detailed spatial fuels information (surface fuel load, crown fuel load, crown to base height, etc.). BIGFOOT uses artificial hotspots as input. Following this approach, the program burns each hotspot once, and the hotspot burns to its <i>estarea</i> value in one time-step (an hour). Based on this, persistence is no longer valid, nor are any of the weighting schemes. Diurnal effects still modify the fire behaviour at each hour, which affects the SFC, CFC and TFC. FGM uses the fire-growth modeling routine. This is experimental and involves the use of various ASCII grids. Many of the inputs are hardcoded to the Lac la Loche fire study. The fire-growth model is currently setup to use only the EMC diurnal approach. [cmc]
MINimum	minimum plume height [m] [-999]
OUTputfilename	The filename (including path) of the CFFEPS Predictions CSV file [output.csv].
RESet	reset time of smoke plume calculations [decimal hours LST] (off if < 0) [-1]
SCR or SMOldering	Smoldering combustion rate [cm/hr] [1.]
SHApe	The choice of fire growth modelling. Options include: • ELLipse (the original elliptical growth method - not recommended); • PERSistence (same as top-hat); • TOPhat (persistence of daily fire growth, with hourly rate constant between 9:00 AM and 9:00 PM, or use THStart and TPEnd);

- **WEIGHted** (persistence of daily growth, with hourly growth that follows an average diurnal trend of rate of spread).
- **WMA** (weighted moving average) uses the hourly calculated value of the Rate of Spread (based on hourly FFMC, etc) over the average of the rate of spread over the past 24 hours.

[weighted]

THStart time to start (exclusive) top-hat fire growth [decimal hours LST]
[9.]

THEnd time to end (inclusive) top-hat fire growth [decimal hours LST]
[21.]

TIMESTEP or
DT or
STEP Model time step [hr]
[1.]

TYPe The type of method used to heat the atmosphere to dry adiabat:

- **AVErage** (heating of average lapse rate to dry adiabat, following methodology used in Anderson *et al.* 2011);
- **DRY** (piecewise integration of upper air profile heated to dry adiabat);
- **LCL** calculates the lifted condensation level as calculated by **WETTER**, passing this as the plume height (note that this is intended for comparative studies and not meant as a prediction of the plume height);
- **PIEcewise** (same as dry);
- **WET** follows the piecewise integration but allows for the latent heat of condensation to be released as into the plume above the lifted condensation level (LCL);
- **WETTER** follows **WET** but incorporates the fuel moisture and water released from combustion in the calculations.

[average]

CFFEPS FILES

The processing cycle of CFFEPS includes information collected by NRCan from the CWFIS and passed on to ECCC at CMC. Next, ECCC reads the file and from GEM creates 72 lines of hourly forecasts for each line of the NRCan file. This is read by **CFFEPS.EXE** and CFFEPS predictions are appended to the end of each line.

The file being passed from one program to the next is a comma-separated variable (CSV) text file. The first line of the file contains a header line of field names, which is read but not used.

The following fields are included in the file. The order of information is strictly followed.

1a. NRCan CFFEPS CSV file

The file created by NRCan contains the input information from the CWFIS. This includes hotspot, weather and fuel information. One line of data is produced per hotspot.

lat	Latitude of the hotspot [decimal degrees]
lon	Longitude of the hotspot [decimal degrees]
rep_date	The report date of the hotspot [YYYY-MM-DD HH:MM]
source	Source of the hotspot data (NASA, NOAA, USFS)
sensor	Sensor (AVHRR, MODIS, VIIRS)
FFMC	Fine Fuel Moisture Code (CFFWIS)
DMC	Duff Moisture Code (CFFWIS)
DC	Drought Code (CFFWIS)
ws	Noon wind speed [km/h]
FWI	Fire Weather Index (CFFWIS)
fuel	Fuel type (CFFBPS)
ROS	Rate of spread (CFFBPS) [m/min]
SFC	Surface fuel consumption (CFFBPS) [kg/m ²]
TFC	Total fuel consumption (CFFBPS) [kg/m ²]
BFC	BORFIRE fuel consumption (no longer used) [kg/m ²]
HFI	Head fire intensity (CFFBPS) [kW/m]
Estarea	Estimated hotspot area [ha]

where CFFWIS is the Canadian Forest Fire Weather Index (FWI) System (Van Wagner 1987) and CFFBPS is the Canadian Forest Fire Behaviour Prediction (FBP) System (Forestry Canada Fire Danger Group 1993). Source and sensor are not used in the CFFEPS model but are there for record keeping. BORFIRE fuel consumption (BFC), also referred to as the Forest Floor Fuel Consumption (FFFC), was a field used in the original BlueSky methodology but not used in CFFEPS (de Groot *et al.* 2009, Pavlovic *et al.* 2016).

1b. Next Generation NRCan CFFEPS CSV File

Development of a new input file for CFFEPS is underway. This file will take advantage of additional information becoming available to the CWFIS. Based on the FBP equations, there are allowances for the following variables (ranked in order of confidence).

All the above fields for the NRCan CFFEPS CSV File (1a) plus the following:

Elev	Elevation [m]	From DEM
Slope	Slope [%]	From DEM
Aspect	Aspect [°]	From DEM
Fuel	FBP fuel type	Broad categories from forest inventory
curing	Grass curing [%]	From MODIS <i>Only applies to grass fuel type (O1)</i>
Greenup	Greenup [y/n]	From MODIS <i>Only applies to deciduous fuel types (D1, M1/2/3/4)</i>
PC	Percent coniferous [%]	Fairly easy to get within +/- 20% with forest inventory <i>Only applies to mixedwood fuel type (M1/2)</i>
SH	Stand height [m]	A very easy metric to derive from forest inventory <i>Only applies to C6 fuel type</i>
CFL	Crown fuel load [kg/m ²]	based on stand height and stem volume
CBH	Crown to base height [m]	Requires more complicated model of species
SD	Stand density [stems/ha]	Not in National Forest Inventory metrics; a complicated combination of age <i>Only applies to C6 fuel type</i>
PDF	Percent dead fir [%]	Hard to get up to date maps of forest pests and impacts <i>Only applies to dead balsam fire fuel type (M3/4)</i>
GFL	Grass fuel load [kg/m ²]	Difficult to derive from NDVI <i>Only applies to grass fuel type (O1)</i>

SFL	Surface fuel load [kg/m ²]	The sum of woody and fine fuels is easier to estimate than the individual FFL and WFL pools
FFL	Fine fuel load [kg/m ²]	Just empirical relationships of age <i>Only applies to C6 and slash fuel type (S1/2/3)</i>
WFL	Woody fuel load [kg/m ²]	Just empirical relationships of age <i>Only applies to C6 and slash fuel type (S1/2/3)</i>

Researchers at NRCan's Northern Forestry Centre are collecting and testing this data for operational use. Whether or not this data is available, CFFEPS is now designed to inset default values when this data is absent.

The new method is currently being tested as part of the Lac La Loche fire. The system uses the new fire progression map technique, which uses hotspot data.

1c. CMC GEM Forecast CSV file

The CMC GEM file builds on the NRCan data set by expanding each line of hotspot data into 72 lines of forecasted hourly data. The following additional forecasted hourly surface and upper air data is appended to each line of the NRCan CFFEPS CSV file.

All the fields for the NRCan CFFEPS CSV File (1a or 1b) plus the following:

UTC	forecast time
temp	temperature [oC or K]
rh	relative humidity [% or fraction]
ZS	surface geopotential height [m or dam]
ws-met	wind speed [km/h or knots]
Precip rate	Precipitation rate [mm/h or m/s]
TS	Surface temperature [oC or K]
T850	850 mb temperature [oC or K]
T700	700 mb temperature [oC or K]
T500	500 mb temperature [oC or K]
T250	250 mb temperature [oC o K]
Z850	850 mb geopotential height [m or dam]
Z700	700 mb geopotential height [m or dam]
Z500	500 mb geopotential height [m or dam]
Z250	250 mb geopotential height [m or dam]

Note that the program is designed to recognize units. The program conducts its calculations in the first listed but if it encounters the latter, it makes a conversion (e.g., if it comes across a temperature greater than 100, it assumes the value is in K and thus converts it to °C).

Also note that the program uses the ICAO standard atmosphere as default values. These may reappear if profile information is not handled properly.

1d. CFFEPS Predictions CSV file

The CFFEPS output file appends fire growth, plume rise, energy and smoke emissions to each line of the CMC GEM Forecast CSV file.

All the fields for the NRCan CFFPES CSV File (1a or 1b) and the CMC GEM Forecast CSV file plus the following:

Area (t)	Cumulative fire area [ha]
Growth (t)	Hourly fire growth [ha]
TotalEmissions	Total emissions [tonnes]
r_smoke	mixing ratio of smoke to clear air [g/kg]
Zplume	Plume height [m]
Mplume	Plume mass [kg]
Qo	Total cumulative energy of the fire [joules]
Qplume	Cumulative energy injected into the plume [joules]
PM F	Hourly particulate matter from flaming combustion
PM S	Hourly particulate matter from smoldering combustion
PM R	Hourly particulate matter from residual combustion

...

NOTE: the last three fields are repeated for every species listed in the emissions CSV file.

2. Emissions Models CSV file

Emissions Model (emodels) CSV file provides a means of linking emissions factors to specific fuel types. Now the file allows for four more fields: **rho2**, **rho4**, **rho6** and **rho8**, used to represent the variables shown on the Table 1. The filename of the Emissions Model CSV file is passed through the CFFEPS input file.

*NOTE: Passing the bulk density field names **rho2**, **rho4**, **rho6** and **rho8** is optional and the values presented in Table 1 reflect the default values (which are used if the fields are missing). Users are encouraged to go with or without all four values for all fuel types; otherwise, a mix of values may produce unpredicted results.*

Fuel	Fuel type (CFFBPS)
Model	Emissions model link to fuel type
rho2	Bulk density 0-2 cm (gm/cm ³)

rho4	Bulk density 2-4 cm (gm/cm ³)
rho6	Bulk density 4-6 cm (gm/cm ³)
rho8	Bulk density 6-8 (+) cm (gm/cm ³)

An example Emissions Model (emodels) CSV file is as follows

```
CWFIS, emodel
01a, 99
01b, 99
01, 99
C1, 99
C2, 99
C3, 99
C4, 99
C5, 99
C6, 99
C7, 99
D1, 99
D2, 99
M1, 99
M2, 99
M3, 99
M4, 99
S1, 99
S2, 99
S3, 99
```

An example of the new Emissions Model (emodels) CSV file is as follows

```
CWFIS, emodel, rho2, rho4, rho6, rho8
01a, 99, 0.05, 0.05, 0.05, 0.05
01b, 99, 0.05, 0.05, 0.05, 0.05
01, 99, 0.05, 0.05, 0.05, 0.05
C1, 99, 0.045, 0.045, 0.045, 0.045
C2, 99, 0.019, 0.034, 0.051, 0.056
C3, 99, 0.015, 0.02, 0.032, 0.066
C4, 99, 0.022, 0.029, 0.045, 0.059
C5, 99, 0.093, 0.093, 0.093, 0.093
C6, 99, 0.03, 0.05, 0.05, 0.05
C7, 99, 0.1, 0.1, 0.05, 0.05
D1, 99, 0.061, 0.061, 0.061, 0.061
D2, 99, 0.061, 0.061, 0.061, 0.061
M1, 99, 0.0265, 0.071, 0.0795, 0.082
M2, 99, 0.0265, 0.071, 0.0795, 0.082
M3, 99, 0.041, 0.061, 0.084, 0.112
```

```

M4, 99, 0.041, 0.061, 0.084, 0.112
S1, 99, 0.2, 0.2, 0.2, 0.2
S2, 99, 0.5, 0.5, 0.3, 0.3
S3, 99, 0.6, 0.6, 1, 1

```

3. Allocation CSV file (optional)

The CFFEPS model allocates fuel consumption into three combustion stages: flaming smoldering and residual. The original calculations followed those described as reduction factors per fuel type in the USFS CONSUME model as shown in Table 2. The filename of the Allocation CSV file is passed through the CFFEPS input file.

NOTE: The use of the Allocation CSV file is optional, and the values represented in Table 2 are the default values.

The allocation file contains the following fields

Fuel	Fuel category being consumed (not FBP fuel type). These include:
-------------	--

SFC_L: Litter layer (top 1.2 cm)
SFC_F: Fermentation layer (1.2-7 cm)
SFC_H: Humus layer (7+ cm)
FFC: Slash - fine fuels
WFC: Slash - woody fuels
GFC: Grass fuel load
CFC_F: Canopy
CFC_B: Branchwood
CFC_S: Snags

GFC is used for O1A/B grass fuel types only
FFC and **WFC** are used form slash fuel type only.
SFC_L, **SFC_F**, **SFC_H** and **CFC_F** are used for the remaining fuel types.
CFC_B and **CFC_S** are not currently used

Flaming	First stage of combustion, lasting 15 minutes
Smoldering	Second stage of combustion
Residual	Final stage of combustion
Remainder	Unburned fuel after combustion
Notes	information on the fuel type (not used)

Note that the flaming stage last 15 minutes while the smoldering and residual stage depend on the depth of burn. These stages are assumed to burn at a rate of 1 cm per hour. Each occur for an equal length of time with the smoldering stage completed before residual stage begins.

An example Allocation CSV file is as follows:

```
Fuel, Flaming, Smoldering, Residual, Remainder, Notes
SFC_L, 0.9, 0.1, 0, 0, Litter layer
SFC_F, 0.1, 0.7, 0.2, 0, Fermentation layer
SFC_H, 0, 0.2, 0.8, 0, Humus layer
FFC, 1, 0, 0, 0, Slash - fine fuels
WFC, 0.4, 0.3, 0.3, 0, Slash - woody fuels
GFC, 0.95, 0.05, 0, 0, Grass fuel load
CFC_F, 0.94, 0.06, 0, 0, Canopy
CFC_B, 0.6, 0, 0, 0.4, Branchwood
CFC_S, 0.25, 0.25, 0, 0.5, Snags
```

4. Emissions CSV file

The emissions CSV file contains emissions factors associated with each desired species as shown in Table 3. The filename of the Emissions CSV file is passed through the CFFEPS input file.

NOTE: the user is given the flexibility to include any species needed but is expected to provide the necessary emissions factors associated with the species (that is, CFFEPS does not provide these values).

Species	Chemical species being modelled
Flaming	Emissions factor of species from flaming combustion [g/kg]
Smoldering	Emissions factor of species from flaming combustion [g/kg]
Residual	Emissions factor of species from flaming combustion [g/kg]
Model	Emissions model link to fuel type

An example Emissions CSV file is as follows

```
Species, Flaming, Smoldering, Residual, Model
PM, 23, 34, 34, 99
PM10, 8.59, 19.63, 19.63, 99
PM2.5, 7.28, 16.63, 16.63, 99
CO, 83, 135, 248, 99
CO2, 1662.33, 1600, 1383, 99
CH4, 3.23, 7.32, 9.94, 99
NMHC, 19.85, 33.87, 56.08, 99
NOX, 1.83, 2.0, 0.45, 99
NH3, 0.99, 1.50, 1.94, 99
SO2, 0.93, 1.06, 1.76, 99
```

5. Diurnal Trend CSV file (optional)

The CFFEPS model employs a diurnal trend in its calculation of fire growth. The original code provided a choice of either a top hat (where the area growth is equal for fixed hours) or a predefined diurnal curve. For example, a fire that has a daily size of 12 hectares (as determined by the hotspot information) would consistently grow 1 hectare per hour following the top-hat approach (assuming it started at 9:00 AM and grew for 12 hours until 9:00 PM). The predefined curve (based on a number of assumptions) would produce a sinusoidal-like growth rate.

The new approach allows the user to define their own hourly diurnal trend. The filename of the Diurnal Trend CSV file is passed through the CFFEPS input file.

NOTE: The use of the Diurnal Trend CSV file is optional.

The diurnal file contains the following fields

Hour	The hour (used an index from 0 to 23)
Weight	The fraction of the daily fire growth used for the current hour

NOTE: the program will make the necessary adjustments so that the weights add up to 1.0 for the entire 24-hour period.

An example Diurnal Trend CSV file is as follows:

```
Hour, Weight
0, 0.015648395
1, 0.012470576
2, 0.010137905
3, 0.008443153
4, 0.004159725
5, 0.003641906
6, 0.004455177
7, 0.005716244
8, 0.017972534
9, 0.032938579
10, 0.043543306
11, 0.055667556
12, 0.065778522
13, 0.078207403
14, 0.090168462
15, 0.104189127
16, 0.101294339
17, 0.091901405
18, 0.072798822
19, 0.057406612
20, 0.044004006
21, 0.033647664
22, 0.025825035
```

23, 0.019983547

6. ASCII grid files

The fire-growth model passes spatial datasets of fuels, topography, etc. as well as input/output ignition grids in the form of ASCII grids. These grids are arranged in column-row order from the lower-left corner. The first six lines preceding the data define the grid size and extents. An example of these lines are as follows:

```
NCOLS 385
NROWS 425
XLLCORNER -903305.8072234492
YLLCORNER 901919.1283280492
CELLSIZE 100
NODATA_VALUE -9999
```

7. PRJ file

The fire-growth model uses PRJ projection files to orient the grid to the real world. CFFEPS uses an old PRJ format as follows

```
Projection      LAMBERT
Zunits          NO
Units           METERS
Spheroid        GRS1980
Xshift          0.0000000000
Yshift          0.0000000000
Parameters
  49 00  0.000 /* 1st standard parallel
  77 00  0.000 /* 2nd standard parallel
-95 00  0.000 /* central meridian
  49 00  0.000 /* latitude of projection's origin
0.00000 /* false easting (meters)
0.00000 /* false northing (meters)
```

PSEUDO CODE FOR CFFEPS

The current CFFEPS code consists of two principal programs and several additional subroutine programs. **CFFEPS.EXE** is the main program that calls the EnergyCalc, PlumeCalc and EmissionsCalc subroutines. Other programs, such as **FWI84.c** and **FBP_2009.c**, are established programs that are not discussed in this document.

Note: this c program is essentially the same as **CFFEPS.c** version 2.02. Slight changes to the code were introduced, essentially making it version 2.03 but this was not formally released. Instead, it was used as a benchmark in the development of version 3.0.

CFFEPS

The program CFFEPS.c contains the main program that runs CFFEPS.

1. Set up the run:
 - a. Read the argument line for the name of the input file. If this is missing, use the default infile.dat.
 - b. Read the FBP defaults (subroutine FBPdefaults found in fbp_2009.c)
 - c. Read the FEPS input file (subroutine ReadFEPSData found in readfeeps.c)
 - d. Define time series arrays for emissions of each pollutant (subroutines ReadEmissionsData and ReadEmissionsModels found in reademissions.c)
 - e. Open CMC GEM forecast CSV file as input. Read in header line.
 - f. Print (to the screen), new header line for the CFFEPS predictions CSV file.
2. Begin reading CFFEPS predictions CSV file:
 - a. Reset arrays to zero.
 - b. Read line of data into buffer. Parse comma separated variables in buffer. Make adjustments for possible variation in units (e.g Kelvin instead of Celsius, etc.)
 - c. Assign upper air values into arrays uP, uZ and uT for future use in piecewise calculation (subroutine PlumeCalcDry found in plume.c)
 - d. Determine if the line read in is a new hotspot (if the latitude, longitude or date does not equal previous line's values). If new, reset arrays to zero.

- e. Reassign FBP fuel type (if required). If there is no matching FBP fuel type, set to non-fuel (NF).
- f. Determine time zone (longitude/15) and parse detection date and time. Adjust detection time from UTC to local time such that noon is at solar zenith. Adjust LDT to LST. Determine UTC and LST time of the forecast record.
- g. Recalculate the BUI value from DMC and DC values. Determine whether the BUI effect is being used (subroutine BUICalc found in fwi84.c)
- h. If new fire, determine the hours from the first hourly CMC forecast to the detection time: if less than zero, the fire starts at the detection size; if less than 24 hours, an initial fire size is calculated (based on persistence routine); if greater than zero, then the fire starts at zero size.
- i. Calculate FFMC from yesterday's value using the technique described in Lawson *et al.* (1996) (subroutine bdl_ffmc found in bdlffmc.c). This is based on the observed noon FFMC. If the time is before noon, yesterday's FFMC is required. This is converged on so that yesterday's FFMC will predict today's observed FFMC (set to 11:59AM, 23:59 hours later).
- j. Calculate FBP values at time of detection (and then at current time) (subroutine FBPCalc found in fbp_2009.c) [Note there are a number of calculations using subroutine ProCalc found in fbp_2009_pro.c. These are done for elliptical growth modelling, which is not being used by FIREWORK.]
- k. Determine discovery time (the number of hours of growth required to reach the detection size). When persistence is being used, this time is assumed to be 24 hours. When elliptical growth modelling is being done, this is solved using ProCalc [see note above].
- l. If current time is prior to discovery, the fire has not yet occurred and does not grow; else:
 - i. If the current time corresponds to the reset time (if being used, not the default), set the area, cumulative energy and total emissions to zero.
 - ii. Determine difference in time between current time and detection time. If less than an hour, calculate partial hour growth; otherwise, calculate hourly growth according to top-hat or weighted approach. Add to total area.
 - iii. Calculate energy of the fire (subroutine EnergyCalc found in energy.c)
 - iv. Calculate the energy released from flaming, smoldering and residual combustion (subroutine EmissionsCalc found in emissions.c). Calculate flaming, smoldering and residual emissions rates in tonnes/h

- v. Distribute the water and the energy over time (subroutine EmissionsOverTime found in emissions.c)
- vi. From the area growth and energy released, calculate the plume rise (subroutine PlumeCalc found in plume.c)
 - (1) if the type of calculation chosen is "dry", "wet" (set Mw to 0) or "wetter" follow a piecewise piecewise integration of upper air profile heated to dry or wet adiabat (subroutine PlumeCalcDry or PlumeCalcWet found in plume.c), otherwise
 - (2) follow the standard average lapse rate method outlined in Anderson *et al.*, 2011 (subroutine PlumeCalc found in plume.c)
 - (a) calculate height based on surface to 850 mb lapse,
 - (b) if recalculated height greater than 2000 m, recalculate height based on surface to 700 mb lapse rate (if resulting height less than 2000, use average),
 - (c) If recalculated height greater than 4000 m, recalculate height based on surface to 500 mb lapse rate (if resulting height less than 4000, use average).
 - (d) if superadiabatic from surface up to 500m, begin recalculate using 850 mb as the base
 - (e) if recalculated height greater than 2000 m, recalculate height based on 850 to 700 mb lapse rate (if resulting height less than 2000, use average)
 - (3) limit height to 250 mb height
 - (4) impose a minimum plume height
- vii. Find emodel that matches the fuel type
- viii. Distribute the emission over time (subroutine EmissionsOverTimeNew found in emissions.c).
- ix. Calculate total emissions and r_smoke
- m. Print results to the output file.
- n. Read next line (go back to step 2. a.)

CFFEPS2

The program CFFEPS2.c is essentially the same program as CFFEPS.c, rewritten in such a way that it calls the subroutine CFFEPScalc (see below). Unfortunately, the file uses 2 in its name, though it is essentially part of version CFFEPS version 3.0.

CFFEPS3

The program CFFEPS3.f90 is essentially the same program as CFFEPS2.c, but written in FORTRAN and calls the subroutine CFFEPScalc (see below).

CFFEPS3.f90 uses the FORTRAN module CFFEPS_Module in CFFEPS_Module.f90 to define the FORTRAN Types, which house the CFFEPS data. It then uses the subroutine readCSV in readCSV.c to read the same CSV of input data used by CFFEPS v 2.02.

CFFEPScalc

The program CFFEPScalc.c contains the subroutine CFFEPScalc that calculates all the CFFEPS model calculations for a given hotspot for a given timestep.

CFFEPScalc can be called from C or from FORTRAN. All information is passed through C structures or FORTRAN types as defined in feps_2011.h and CFFEPS_Module.f90, respectively.

The program CFFEPS3.f90 serves as an example of how CFFEPScalc can be called from FORTRAN. Note there is a small subroutine cffepscal_ at the bottom of the file (lower case letter and with a "_" at the end), which is the subroutine that FORTRAN actually calls. In turn, cffepscal_ calls CFFEPScalc.

Note: this subroutine constitutes CFFEPS version 3.0. The subroutine is called by both CFFEPS2.c and CFFEPS3.f90. Extensive testing was done running CFFEPS.c (version 2.02), CFFEPS2.c and CFFEPS3.f90 to ensure they were producing similar results (due to the evolution of the code, there were slight differences between CFFEPS v 2.02 and the current version CFFEPS v 3.0, whether c or f90).

1. Assign weight[] values to top-hat approach (if used)
2. Reassign FBP fuel type (if required). If there is no matching FBP fuel type, set to non-fuel (NF).
3. Determine time zone (longitude/15). Adjust detection and forecast (CMC) times from UTC to local time such that noon is at solar zenith. Adjust LDT to LST (hardcoded to LST). Determine Julian dates.
4. Recalculate the BUI value from DMC and DC values. Determine whether the BUI effect is being used (subroutine BUICalc found in fwi84.c)

5. If new fire, determine the hours from the first hourly CMC forecast to the detection time: if less than zero, the fire starts at the detection size; if less than 24 hours, an initial fire size is calculated (based on persistence routine); if greater than zero, then the fire starts at zero size.
6. Calculate FFMC from yesterday's value using the technique described in Lawson *et al.* (1996) (subroutine bdl_ffmc found in bdlffmc.c). This is based on the observed noon FFMC. If the time is before noon, yesterday's FFMC is required. This is converged on so that yesterday's FFMC will predict today's observed FFMC (set to 11:59AM, 23:59 hours later).
7. Calculate FBP values at time of detection (and then at current time) (subroutine FBPCalc found in fbp_2009.c) [Note there are a number of calculations using subroutine ProCalc found in fbp_2009_pro.c. These are done for elliptical growth modelling, which is not being used by FIREWORK.]
8. Determine discovery time (the number of hours of growth required to reach the detection size). When persistence is being used, this time is assumed to be 24 hours. When elliptical growth modelling is being done, this is solved using ProCalc [see note above].
9. If current time is prior to discovery, the fire has not yet occurred and does not grow; else:
 - a. If the current time corresponds to the reset time (if being used, not the default), set the area, cumulative energy and total emissions to zero.
 - b. Determine difference in time between current time and detection time. If less than an hour, calculate partial hour growth; otherwise, calculate hourly growth according to top-hat or weighted approach. Add to total area.
 - c. Calculate energy of the fire (subroutine EnergyCalc found in energy.c)
 - d. Calculate the energy released from flaming, smoldering and residual combustion (subroutine EmissionsCalc found in emissions.c). Calculate flaming, smoldering and residual emissions rates in tonnes/h
 - e. Distribute the energy over time (subroutine EmissionsOverTime found in emissions.c)
 - f. From the area growth and energy released, calculate the plume rise (subroutine PlumeCalc found in plume.c)
 - i. if the type of calculation chosen is "dry", follow a piecewise piecewise integration of upper air profile heated to dry adiabat (subroutine PlumeCalcDry found in plume.c), otherwise
 - ii. follow the standard average lapse rate method outlined in Anderson *et al.*, 2011 (subroutine PlumeCalc found in plume.c)

- (1) calculate height based on surface to 850 mb lapse,
- (2) if recalculated height greater than 2000 m, recalculate height based on surface to 700 mb lapse rate (if resulting height less than 2000, use average),
- (3) If recalculated height greater than 4000 m, recalculate height based on surface to 500 mb lapse rate (if resulting height less than 4000, use average).
- (4) if superadiabatic from surface up to 500m, begin recalculate using 850 mb as the base
- (5) if recalculated height greater than 2000 m, recalculate height based on 850 to 700 mb lapse rate (if resulting height less than 2000, use average)

iii. limit height to 250 mb height

iv. impose a minimum plume height

g. Find emodel that matches the fuel type

h. Distribute the emission over time (subroutine EmissionsOverTimeNew found in emissions.c).

i. Calculate total emissions and r_smoke

CFFEPS_Module

The FORTRAN module CFFEPS_Module in CFFEPS_Module.f90 defines the FORTRAN types to match the c structures used by CFFEPScalc, as defined in feps_2011.h.

fire_propagate

The program fgmgrid.c contains the subroutine fire_propagate. This routine propagates a fire through a regular grid until the fire reaches the edge or until the simulation time expires. A number of spatial grids are passed to fire_propagate as pointers to ASCII grids. As well, the direction of growth (forward or backward), the method (8 or 16-point growth), and the standard feps, fbp and fgmwx inputs data structures are passed.

1. Define grid cell relations to propagation spread directions: 8 and 16 point models.

2. Create 2 stacks (OldStack and NewStack) to track locations of cells that have changed.
3. Build a grid to track steps when cells have last changed.
4. While cells continue to change
 - a. Step through the old stack of changed cells
 - i. Unless the edge of the grid has been reached or time has expired
 - ii. Determine Fuel types of origin
 1. Determine Fuel types of surrounding targets
 2. Calculate time to burn out of origin cell into target cells
 3. If ignition time in target cell is more recent than the previously calculated time, consider it a new ignition and add it to NewStack and count of changed values
 - b. Transfer newly changed locations to old stack

In step 4.a.ii.2, fire_propagate calls another routine burn_cell that interpolates weather, calculates the FBP rates of spread and then the amount of time to burn from the center of one cell to the next.

EmissionsCalc

The program Emissions.c contains the subroutine EmissionsCalc used to estimate the amounts of flaming, smoldering and residual fuel consumptions used by CONSUME 3.0.

1. Apply the average bulk density per fuel type to bulk densities for 2, 4 6 and 8 cm depths.
2. Overwrite these values with depth specific values for certain fuels.
3. Calculate the bulk densities of the L (0-1.2 cm), F (1.2-7 cm) and H (7-18 cm) layers from the 2 cm depth data. Water mass (Mw) is calculated from the moisture content as described by the FPMC, (top 2 cm), DMC (2-5 cm), DC (5+ cm) and the depth of burn
 - a. If grass fuel type (O1), burn off the entire grass fuel load. Use the grass allocation values for flaming, smoldering and residual combustion (see Table 2).

- b. else, if slash, use the total fuel consumption (TFC). Use the slash allocation values for flaming, smoldering and residual combustion (see Table 2).
 - c. else, burn off the LFH layers in sequence until the entire SFC is accounted for. Then use the crown fuel consumption (CFC) for any canopy burned. Use the corresponding allocation values for flaming, smoldering and residual combustion (see Table 2). Calculate water released [kg/m²], including moisture released due to fuel moisture (mcL, mcF, mcH) and for chemical decomposition, required for the subroutine PlumeCalcWet
 4. Calculate depth of burn [cm] per stage
 5. Calculate energy release per stage ((Qflaming, Qsmoldering, Qresidual).
-

EmissionsOverTime

The program Emissions.c contains the subroutine EmissionsOverTime used to distribute energy, fuel consumption, and smoke emissions over time for a species.

The CFFEPS program creates pointers to arrays to represent the time series of smoke emissions. An array is created for each chemical species and each array element represents one-time step in CFFEPS. During each time step in CFFEPS, new flaming, smoldering and residual fuel consumptions are calculated in EmissionsCalc, then EmissionsOverTime is called to distribute the emissions into the arrays (one array per chemical).

1. Calculate the total residence time in timeteps for flaming, smoldering and residual stages
 2. Calculate the emissions rates [tonnes/hr] when jModel >= 0
 - a. if jModel == -1, use this technique to come up with energy per hour
 - b. if jModel == -2, use this technique to come up with water released per hour, required for the subroutine PlumeCalcWet
 - c. if jModel < -2, use this technique to come up with fuel consumed per hour.
 3. Stepping through each timestep,
 - a. calculate partial residence times remaining in decimal hours
 - b. calculate the emissions for the partial hour (recalculated from hours to timesteps)
-

EnergyCalc

The program Energy.c contains the subroutine EnergyCalc. Parameters are passed to EnergyCalc through the structures FEPS and FBP. These structures are filled through information passed to CFFEPS through the input file.

1. Units are converted to SI units.
2. Determine fire area, growth, perimeter and radius.
3. Assign default duff characteristics by fuel type based on the literature and the fuel moisture based on the FPMC, DMC and DC.
 - a. If grass fuel type (O1), burn off the entire grass fuel load.
 - b. Else, if slash, use the total fuel consumption (TFC). Use the slash reduction values for flaming, smoldering and residual combustion (see Table 2).
 - c. Else, burn off the LFH layers in sequence until the entire SFC is accounted for. Then use the crown fuel consumption (CFC) for any canopy burned. Use the corresponding reduction values for flaming, smoldering and residual combustion (see Table 2).
4. Calculate depth of burn and the duff consumption from the surface fuel consumption (SFC) and densities
5. Water mass is calculated from the moisture content as described by the FPMC, (top 2 cm), DMC (2-5 cm), DC (5+ cm) and the depth of burn. Add the water mass from the crown fuel consumption (TFC-SFC) and the foliar moisture content (FMC).
6. Calculate Q_{fire} , Q_w , Q_f , Q_s , Q_r , Q_{inc} and from these Q_{plume}

Note: the energy calculations are not finalized and there are a few alternate means of calculating these. For the current radiation model, Q_r is based on the ratio of the surface area of the flame wall to the size of the fire. For the current incomplete combustion model, Q_{inc} is based on Q_f alone (multiplied by $CFB/2$).

PlumeCalc

The program Plume.c contains the subroutine PlumeCalc. This routine calculate plume rise using an average lapse rate for the atmosphere. Parameters are passed to PlumeCalc through the structures FEPS and FBP. These structures are filled through information passed to CFFEPS through the input file.

1. If the environmental lapse rate is unstable ($\leq -9.80\text{C/km}$), or if the area or Q_{plume} is less than zero, the plume height is set to -9999 (missing) and the subroutine is exited, else
 2. Units are converted to SI units.
 3. Q_{plume} is the energy injected into the plume as calculated by EnergyCalc.
 4. If past energy Q_0 present, add it to Q_{plume} to get Q_t .
 5. Starting with an estimate height of 1000 m and converge on the plume height.
 - a. Calculate the energy per unit mass q needed to heat plume to dry adiabat
 - b. Calculate the column mass of the plume M
 - c. Calculate air density in plume
 - d. Adjust mass for entrainment
 - e. Calculate total energy needed to heat plume to dry adiabat
 - f. If energy greater than Q_t , halve the height-step size and reduce the height, otherwise increase the height.
 - g. Repeat calculations until the height-step size is less than a metre.
 6. From the previous plume energy Q_0 , calculate the black body radiation loss Q_{bb} .
-

PlumeCalcDry

The program Plume.c contains the subroutine PlumeCalcDry. This routine calculate plume rise by conducting a piecewise integration of energy through the reported layers for the atmosphere. Parameters are passed to PlumeCalcDry through the structures FEPS and FBP. These structures are filled through information passed to CFFEPS through the input file.

1. If the area or Q_{plume} is less than zero, the plume height is set to -9999 (missing) and the subroutine is exited , else
2. Units are converted to SI units.
3. Q_{plume} is the energy injected into the plume as calculated by EnergyCalc.
4. If past energy Q_0 present, add it to Q_{plume} to get Q_t .
5. Set array pointers to first upper air data point.
6. Set cumulative energy per unit mass, cumulative mass and plume height to zero.
7. Begin converging on plume height.
 - a. If stepping upwards ($Q_{\text{atm}} < Q_t$), read in next level of data
 - i. Level 2 is the upper air data for the current level,
 - ii. Level 1 is the upper air data for the next level
 - b. Calculate energy required to heat atmosphere to dry adiabat. Solution is found by calculating energy contained in trapezoid defined by (T1, theta1), (T2, theta2), (T3, theta3), (T4, theta4) where the temperatures are at the next level (T1), the current level (T2), while T3 and T4 are those when lowered adiabatically to the surface (T1 to T4, T2 to T3)
 - i. Calculate T3 and T4 using dry adiabat,
 - ii. Calculate potential temperatures,
 - iii. If stepping upward, q_c and M_c are updated (when stepping down, q_c and M_c remain constant),
 - iv. Calculate area in trapezoid (new strategy, subtracting triangles to calculate area in trapezoid),
 - v. Add new energy and mass to previous values q_c and M_c .
 - c. Calculate air density in plume.
 - d. Adjust mass for entrainment.
 - e. If energy required to heat the air mass (Q_{atm}) is greater than plume energy (Q_t), begin calculations downward until $Q_{\text{atm}} < Q_t$ again, in which case the solution is reached.
8. From the previous plume energy Q_0 , calculate the black body radiation loss Q_{bb} .

PlumeCalcWet

The program Plume.c contains the subroutine PlumeCalcWet. This routine calculate plume rise by conducting a piecewise integration of energy through the reported layers for the atmosphere. In addition, if the plume is predicted to rise above the liufted condinsation level, condensation is included in the calculations. Parameters are passed to PlumeCalcWet through the structures FEPS and FBP. These structures are filled through information passed to CFFEPS through the input file.

1. If the area or Q_{plume} is less than zero, the plume height is set to -9999 (missing) and the subroutine is exited, else
2. Units are converted to SI units.
3. Q_{plume} is the energy injected into the plume as calculated by EnergyCalc.
4. If past energy Q_0 present, add it to Q_{plume} to get Q_t .
5. Set array pointers to first upper air data point.
6. Set cumulative energy per unit mass, cumulative mass and plume height to zero.
7. Begin converging on plume height.
 - a. If stepping upwards ($Q_{\text{atm}} < Q_t$), read in next level of data
 - i. Level 2 is the upper air data for the current level,
 - ii. Level 1 is the upper air data for the next level
 - b. Calculate energy required to heat atmosphere to dry adiabat. Solution is found by calculating energy contained in trapezoid defined by (T_1, θ_1) , (T_2, θ_2) , (T_3, θ_3) , (T_4, θ_4) where the temperatures are at the next level (T_1), the current level (T_2), while T_3 and T_4 are those when lowered adiabatically to the surface (T_1 to T_4 , T_2 to T_3)
 - i. Calculate T_3 and T_4 using dry adiabat,
 - ii. Calculate potential temperatures,
 - iii. If stepping upward, q_c and M_c are updated (when stepping down, q_c and M_c remain constant),
 - iv. Calculate area in trapezoid (new strategy, subtracting triangles to calculate area in trapezoid),
 - v. Add new energy and mass to previous values q_c and M_c .
 - c. Calculate air density in plume.

- d. Adjust mass for entrainment.
 - e. Adjust mixing ration for moisture released due to fuel moisture and for chemical decomposition (if $M_w > 0$)
 - f. an approximation of condensation
 - g. If energy required to heat the air mass (Q_{atm}) is greater than plume energy (Q_t), begin calculations downward until $Q_{atm} < Q_t$ again, in which case the solution is reached
8. From the previous plume energy Q_o , calculate the black body radiation loss Q_{bb} .
-

readCSV

The program `readCSV.c` contains the subroutine `readCSV`, which reads in the GEM input file.

Originally part of the `CFFEPS.c` program, `readCSV` was designed as a conduit to FORTRAN, enabling `CFFEPSv3.f90`, a test FORTRAN program, a chance to read in the same input data as the C code, preparing it for a call to `CFFEPScalc`.

Parameters are passed to `PlumeCalcDry` through the structures `FEPS` and `FBP`. These structures are filled through information passed to `CFFEPS` through the input file.

REFERENCES

- Anderson, G. K., Sandberg, D. V., Norheim, R. A. 2004. Fire Emission Production Simulator (FEPS) User's Guide. USDA Forest Service, 99.
- Anderson, K.R. 2009. A comparison of hourly Fine Fuel Moisture Code calculations within Canada. In 8th Symp. on Fire and Forest Meteorology, Oct 13-15, 2007. Kalispell, MT. Am. Meteorol. Soc., Boston, MS.
- Anderson, K.R. 2000. Incorporating smoldering into fire growth modelling. Pages 31-36 in 3rd Symp. on Fire and Forest Meteorology, Jan. 9-14, 2000, Long Beach, CA. Am. Meteorol. Soc., Boston, MS.
- Anderson, K.R., Englefield, P.E. 2001. Quantile characteristics of forest fires in Saskatchewan. Pages 9-16 in 4rd Symp. on Fire and Forest Meteorology, Nov. 12-15, 2001, Reno, NV. Am. Meteorol. Soc., Boston, MS.
- Anderson, K.R., Pankratz, A. 2011. A thermodynamic approach to estimating smoke plume heights. In 9th Symp. on Fire and Forest Meteorology, Oct 18-20, 2011. Palm Springs, CA. Am. Meteorol. Soc., Boston, MS.
- Anderson, K.; Pankratz, A.; Mooney, C.; Fleetham, K. 2018. The Alberta smoke plume observation study. *Earth System Science Data* 10(1):325-337.
- Anderson, K.R.; Pankratz, A. Mooney, C.; Fleetham, K. 2015. An Alberta Plume Study and Comparison to Predictive Models. In 11th Symp. on Fire and Forest Meteorology, May 04 - 07, 2015. Minneapolis, MN. Am. Meteorol. Soc., Boston, MS. 11 pp.
- Beck, J.A. and Armitage, O.B. 2004. Diurnal fine fuel moisture characteristics at a northern latitude. Pages 211-221 in 22nd Tall Timbers Fire Ecology Conference: Fire in Temperate, Boreal and Montane Ecosystems, October 15-18, 2001, Kananaskis, Alberta. Tall Timbers Research Station, Tallahassee, FL.
- Chen, J.; Anderson, K.; Pavlovic, R.; Moran, M.D., Englefield, P., Thompson, D.K., Munoz-Alpizar, R; Landry, H. 2019. The FireWork v2.0 air quality forecast system with biomass burning emissions from the Canadian Forest Fire Emissions Prediction System v2.03. *Geosci. Model Dev.*, 12, 3283–3310.
- Côté, J., Gravel, S., M,thot, A., Patoine, A., Roch, M., Staniforth, A. 1998. The operational CMC MRB Global Environmental Multiscale (GEM) model: Part I Design considerations and formulation. *Monthly Weather Review* 126, 1373B1395.
- de Groot, W.J., Pritchard, J.M., Lynham, T.J. 2009. Forest floor fuel consumption and carbon emissions in Canadian boreal forest fires. *Can. J. For. Res.* 39: 367-382.

- Englefield, P., Lee, B.S., Fraser, R.H., Landry, R., Hall, R.J., Lynham, T.J., Cihlar, J., Li, Z., Jin, J., Ahern, F.J. 2004. Applying geographic information systems and remote sensing to forest fire monitoring, mapping and modelling in Canada. In Proceedings of the 22nd Tall Timbers Fire Ecology Conference: Fire in Temperate, Boreal and Montane Ecosystems, Kananaskis, Alberta. Tall Timbers Research Station: Tallahassee, FL
- Forestry Canada Fire Danger Group. 1992. Development and structure of the Canadian Forest Fire Behavior Prediction System. Inf. Rep ST-X-3. Forestry Canada, Science and Sustainable Development Directorate, Ottawa, Ont., 63 pp.
- Fraser, R.H.; Cihlar, J. 2000. Hotspot and NDVI differencing synergy (HANDS): a new technique for burned area mapping over boreal forest. *International Journal of Remote Sensing* 74(3):362–376.
- Griffin, D., Sioris, C., Chen, J., Dickson, N., Kovachik, A., De Graaf, M., Nanda, S., Veefkind, P., Dammers, E., McLinden, C.A. and Makar, P., 2020. The 2018 fire season in North America as seen by TROPOMI: aerosol layer height intercomparisons and evaluation of model-derived plume heights. *Atmospheric Measurement Techniques*, 13(3): 1427-1427.
- Irabarne, J.V.; Godson, W.L. 1973. *Atmospheric Thermodynamics*. D. Reidel Publ. Co., Dordrecht, NL. 222 p.
- Kourtz P.; Nozaki S.; O'Regan WG. 1977. Forest Fires in the Computer – a Model to Predict the Perimeter Location of a Forest Fire. Can. For. Serv., For. Fire Res. Inst., Inf. Rep. FF-X-65. Petawawa, Ont.
- Larkin, N.K., O'Neill, S.M., Solomon, R., Raffuse, S., Strand, T., Sullivan, D.C., Krull, C., Rorig, M., Peterson, J., Ferguson, S.A., 2010. The BlueSky smoke modeling framework. *International Journal of Wildland Fire*, 18(8), pp.906-920.
- Landry, C., Parent, R., Deschênes, J.-F., Giguère, A., Hardy, G., Verret, R. 2005. Operational Scribe nowcasting sub-system: objective verification results. In 21st International Conference on Interactive Information Processing Systems (IIPS) for Meteorology, Oceanography, and Hydrology, San Diego, CA. Am. Meteorol. Soc., Boston, MS.
- Lawson, B.D., Armitage, O.B., Hoskins, W.D. 1996. Diurnal Variation in the Fine Fuel Moisture Code: Tables and Computer Source Code. "Canada-British Columbia Partnership Agreement on Forest Resource Development: FRDA II." Can. For. Serv./B.C. Min. of Forestry, Victoria, B.C. FRDA report 245.
- Nadeau, L.B., McRae, D.J., Jin, J.-Z. 2005. Development of a national fuel-type map for Canada using fuzzy logic. Information Report NOR-X-406. Can. For. Serv., Edmonton, AB)
- Pavlovic, R., Chen, J., Anderson, K., Moran, M. D., Beaulieu, P. A., Davignon, D., Cousineau, S. 2016. The FireWork air quality forecast system with near-real-time biomass burning emissions: Recent developments and evaluation of performance for the 2015 North

- American wildfire season. *Journal of the Air & Waste Management Association*, 66(9), 819-841.
- Pouliot, G.; Pierce, T.; Benjey, W.; O'Neill, S.M.; Ferguson, S.A. 2005. Wildfire emission modeling: integrating BlueSky and SMOKE. In '14th International Emission Inventory Conference', 11–14 April 2005, Las Vegas, NV.
- Prichard, S.J., Ottmar, R.D., Anderson, G.K. 2006. *Consume 3.0 User's Guide*. Seattle, WA: USDA Forest Service, Pacific Northwest Research Station, Pacific Wildland Fire Sciences Laboratory, Fire and Environmental Research Applications Team. 234 p.
- Quayle, B., Lannom, K., Finco, M., Norton, J., Warnick, R. 2003. Monitoring wildland fire activity on a national-scale with MODIS imagery. In 2nd International Wildland Fire Ecology and Fire Management Congress and 5th Symposium on Fire and Forest Meteorology. Am. Meteorol. Soc., Boston, MS.
- Stieb, D., Yao, J., Henderson, S., Pinault, L., Smith-Doiron, M.H., Robichaud, A., van Donkelaar, A., Martin, R.V., Menard, R., Brook, J. R. 2019. Variability in ambient ozone and fine particle concentrations and population susceptibility among Canadian health regions, *C. J. Public Health* 110 (2): 149-158.
- Stocks, B.J., Lawson, B.D., Alexander, M.E., Van Wagner, C.E., McAlpine, R.S., Lynham, T.J., Dub, D.E. 1989. The Canadian Forest Fire Danger Rating System: an overview. *Forestry Chronicle* 65: 450-457.
- Van Wagner, C.E. 1987. Development and structure of the Canadian Forest Fire Weather Index System. *Can. For. Serv., Ottawa, Ont. For. Tech. Rep.* 35. 37 pp.
- Van Wagner, C.E. 1977. A Method of Computing Fine Fuel Moisture Content throughout the Diurnal Cycle. *Can. For. Serv., Chalk River, Ont. Inf. Rep.* PS-X-69. 15 pp.
- Verret, R., Vigneux, D., Marcoux, J., Petrucci, F., Landry, C., Pelletier, L., Hardy, G. 1997. SCRIBE 3.0 A Product Generator. Pp. 392-395 in 13th International Conference on Interactive Information and Processing Systems for Meteorology, Oceanography and Hydrology, Long Beach, CA. Am. Meteorol. Soc., Boston, MS.
- Wilson, L.J., Vallée, M. 2002. The Canadian Updateable Model Output Statistics (UMOS) system: design and development tests. *Weather and Forecasting* 17, 206B222.
- Wilson, L.J., Vallée, M. 2003. The Canadian Updateable Model Output Statistics (UMOS) system: validation against perfect prog. *Weather and Forecasting* 18, 288B302.

APPENDIX A: DETAILED *PLUMEWET* CALCULATIONS

The new plume-rise routine (*PlumeWet*) addresses condensation in the plume. These calculations are based on heating the layers from the surface to the lifted condensation level (LCL) to a dry adiabat and then to a pseudo-adiabat above the LCL.

A sample profile is shown: blue is the dry-bulb temperature, green is the wet-bulb temperature. Consider the energy required for the plume to rise from level 2 and level 1. This assumes the calculations below level 2 are complete and that the air mass below level 2 has already been heated to a dry adiabat.

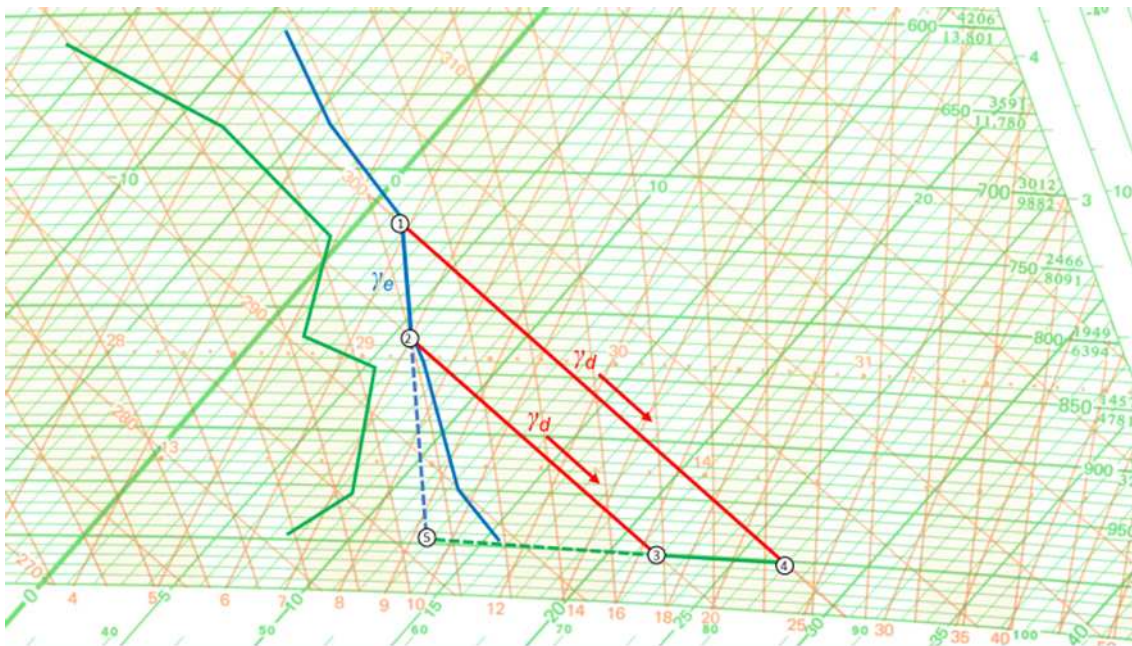


Figure A.16. Sample profile.

The energy is defined as that required to heat the air between the two levels to a dry adiabat (Fig. A. 1). Conditions at points 1 and 2 are lowered along dry adiabats to surface reference points 3 and 4. Point 5 is a reference point used to simplify calculations. It is worth noting that these temperatures T_3 and T_4 are much warmer than the observed environmental surface temperature. This reflects the impact of the energy of the fire injected into the plume and then mixed and does not represent the temperature of the fire.

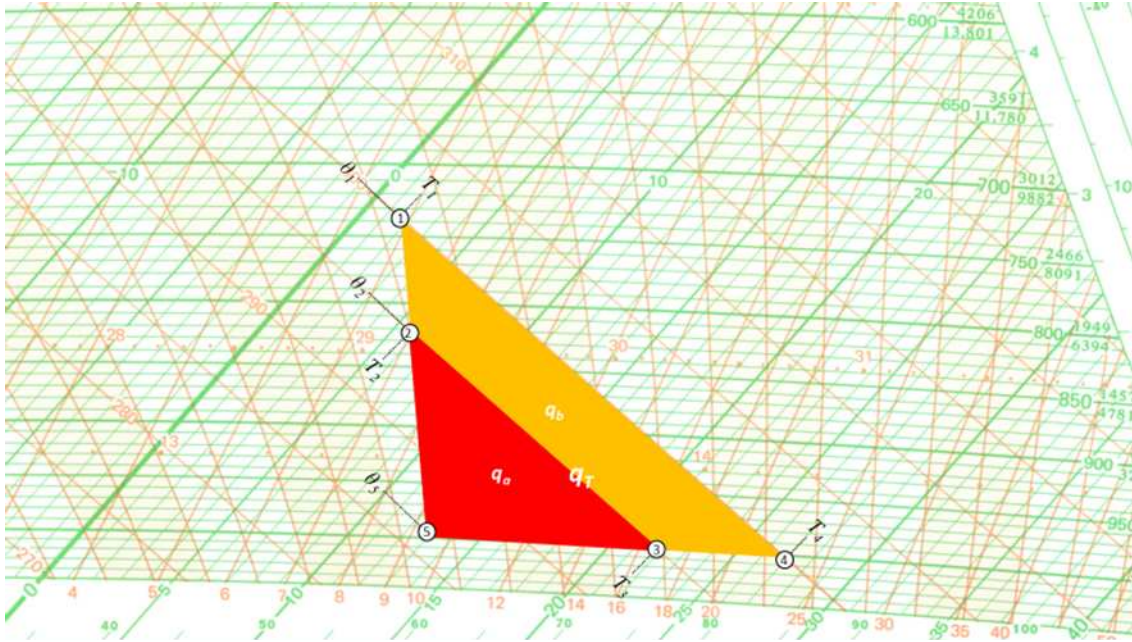


Figure A.17. Energy diagram

Area on a tephigram (Fig. A.2) is proportional to energy. Thus, triangles 2-3-4 (q_a) and 1-4-5 ($q_T = q_a + q_b$) are calculated. The difference of the two triangles give the area in trapezoid 1-2-3-4 (q_b), which is the answer.

$$q_T = -0.5c_p \ln (\theta_1/\theta_5)(T_1 - T_4)$$

$$q_a = -0.5c_p \ln (\theta_2/\theta_5)(T_2 - T_3)$$

$$q_b = q_T - q_a$$

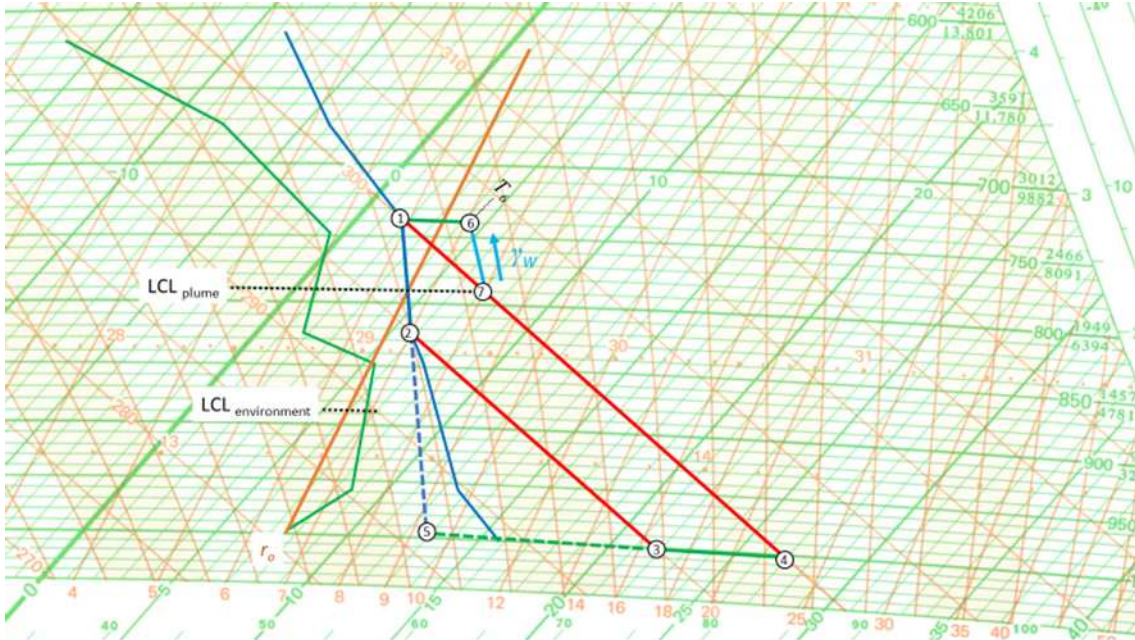


Figure A.18. Sample profile with LCL.

Returning to the profile (Fig. A.3), this layer crosses the lifted condensation level (LCL) as defines by mixing ratio r_o at the surface. Note that the heating of the air inside the plume, the LCL_{plume} is higher than $LCL_{environment}$ outside the plume.

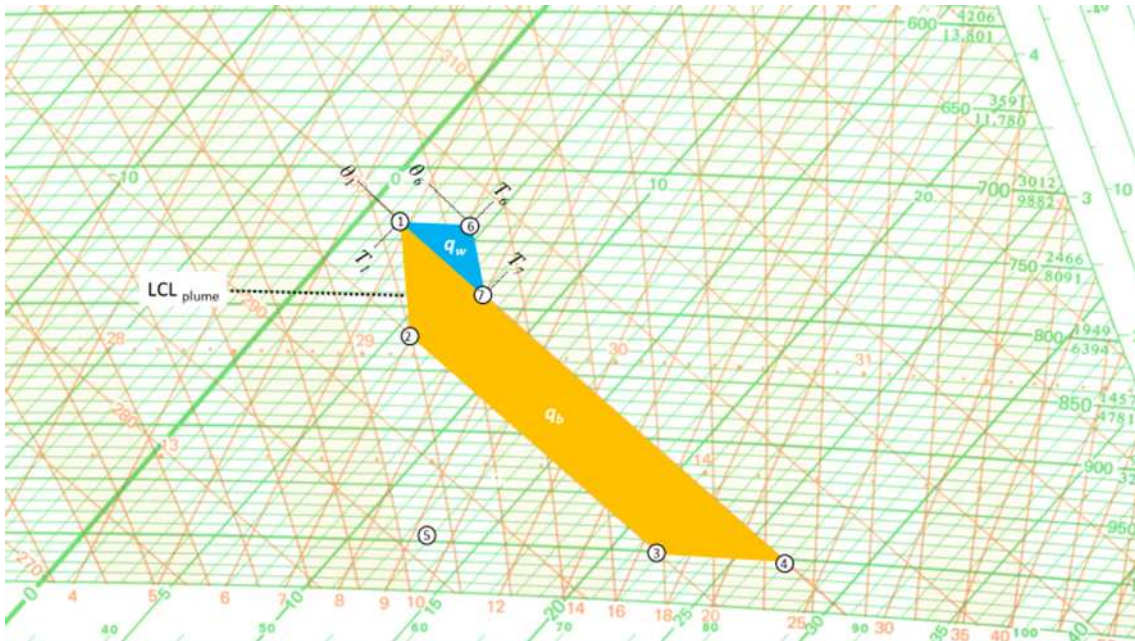


Figure A.19. Energy released due to condensation.

Returning to the energy diagram (Fig. A.4), a new point, T_7 , is the temperature of the plume at the LCL defined by the temperature lifted dry adiabatically from point 4 to the LCL.

From T_7 , a new point, T_6 , is defined as the temperature reached when following the pseudo-adiabat to the top of the layer 1. Note that the pseudo-adiabat is approximated as a straight line from the LCL at point 7 (so the routine is less reliable over thick layers).

This creates a new energy triangle 1-6-7 resulting from the latent heat of condensation released into the plume.

$$q_w = -0.5c_p \ln(\theta_6/\theta_1)(T_1 - T_7)$$

This energy is added to the energy of the fire providing more buoyant lift in the plume.

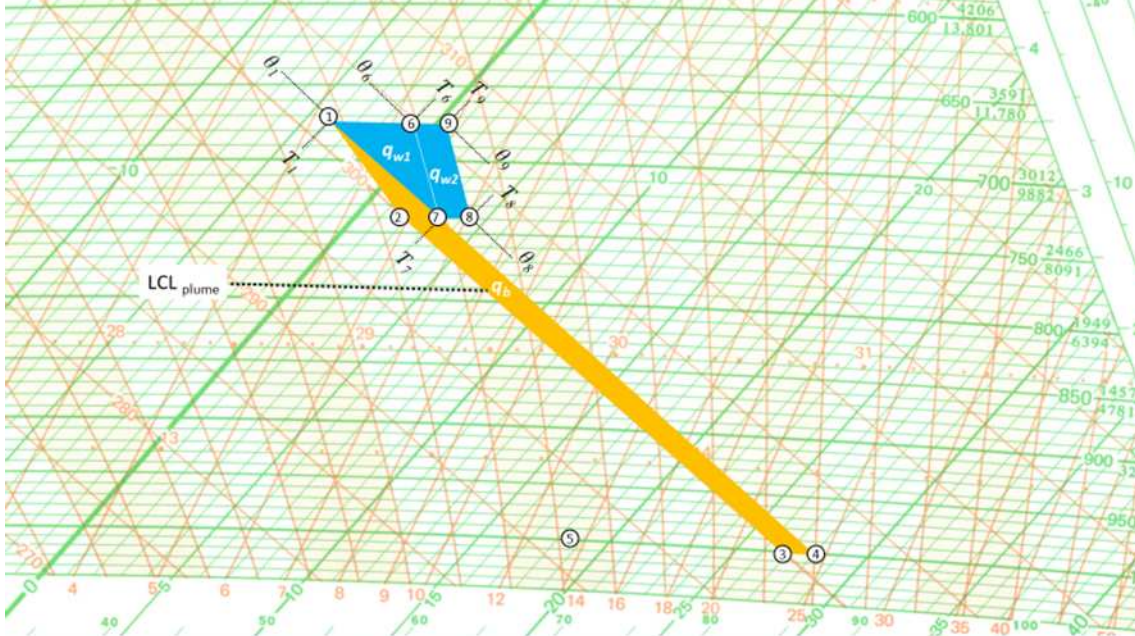


Figure A.20. Condensation at levels above LCL.

Layers above the LCL require additional calculations (Fig. A.5), setting T_8 to the previous T_6 and calculating the energy released in areas q_{w1} and q_{w2} .

$$q_w = q_{w1} + q_{w2}$$

$$q_w = -0.5c_p \ln(\theta_6/\theta_1)(T_1 - T_7) - c_p \ln(\theta_9/\theta_6)(T_6 - T_7)$$

In summary, by including condensation to the plume-rise routine, we expect to see higher plume. This will occur only if the plume reaches the lifted condensation level as defined by temperature inside the plume (and not the outside environment). This process effectively describes that which occurs in a pyrocumulus.